

Structure of the human CTF18–RFC clamp loader bound to PCNA

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eLife Assessment

This **important** study resolves the structure of one missing piece of the eukaryotic DNA replication fork, the leading strand clamp loader. Overall, the data are **convincing**, with electron microscopy data providing a strong basis for analyzing differences and similarities with other RFC complexes. A minor point is that the evidence supporting the proposed role of the β -hairpin is **incomplete**.

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Abstract

Sliding clamps like PCNA are crucial processivity factors for replicative polymerases, requiring specific clamp loaders for loading onto DNA. The human alternative clamp loader CTF18-RFC interacts with the leading strand polymerase Pol ϵ and loads PCNA onto primer/template DNA using its RFC pentameric module. Here, we provide a structural characterization of the human CTF18-RFC complex and its interaction with PCNA. Our cryo-EM data support that the Ctf8 and Dcc1 subunits of CTF18-RFC, which form the regulatory module interacting with Pol ϵ , are flexibly tethered to the RFC module. A 2.9 Å cryo-EM structure shows the RFC module bound to PCNA in an auto-inhibited conformation similar to the canonical RFC loader, marking the initial step of the clamp-loading reaction. The unique RFC1 (Ctf18) large subunit of CTF18-RFC, which based on the cryo-EM map shows high relative flexibility, is anchored to PCNA through an atypical low-affinity PIP box in the AAA+ domain and engages the RFC5 subunit using a novel β -hairpin at the disordered N-terminus. We show that deletion of this β -hairpin impairs the CTF18-RFC–PCNA complex stability, slows down clamp loading, and decreases the rate of primer synthesis by Pol ϵ . Our research identifies distinctive structural characteristics of the human CTF18-RFC complex, providing insights into its role in PCNA loading and the stimulation of leading strand synthesis by Pol ϵ .

Introduction

High fidelity and efficiency are critical for chromosomal DNA duplication in eukaryotes. This precision is achieved through the coordinated action of replicative DNA polymerases, Pol δ and Pol ϵ , anchored to the sliding clamp, Proliferating Cell Nuclear Antigen (PCNA) (1). The ring-shaped PCNA homotrimer encircles DNA, enhancing the processivity of these polymerases in DNA replication and repair processes, including excision repair and homologous recombination (2), and plays an important role in sister chromatid cohesion establishment (3). The closed-ring topology of PCNA necessitates the transient opening of one interface to facilitate the incorporation and trapping of primer-template DNA. This critical function is executed by pentameric AAA+ (Adenosine Triphosphatase Associated) ATPases, collectively known as clamp loaders (4–6). Extensive structural and biochemical work in different organisms (7–12) established a common mechanism for the clamp-loading reaction, which proceeds through three main steps (13–18). Initially, the clamp loader engages with the intact clamp ring, adopting an autoinhibited configuration that precludes DNA binding (10). This stage sets the groundwork for the subsequent transition, where the clamp loader alters its conformation to an activated state. This change facilitates the opening of the clamp ring, creating a gap sufficiently large to permit the passage of duplex DNA and aligning the clamp loader for primer-template DNA interaction. The final phase of this process is initiated by the binding of DNA to the clamp loader, catalysing ATP hydrolysis. DNA binding prompts the closure of the clamp and hydrolysis of ATP induces the concurrent disassembly of the closed clamp loader from the sliding clamp-DNA complex, completing the cycle necessary for the engagement of the replicative polymerases to start DNA synthesis.

The canonical Replication Factor C (RFC), consisting of a large RFC1 subunit and four subunits (RFC2, RFC3, RFC4, and RFC5), acts as the primary clamp loader for PCNA (4, 19). However, eukaryotes also employ alternative loaders (20), including CTF18-RFC (6, 21–24), which likely use a conserved loading mechanism but are functionally specialized through specific protein interactions and context-dependent roles in DNA replication.

Genome-wide analyses revealed that CTF18-RFC predominantly facilitates PCNA loading on the leading strand, in contrast to the canonical RFC complex, which mainly loads PCNA on the lagging strand (25). Although CTF18-RFC is not essential for bulk DNA replication in yeast (26, 27), it plays a vital role in ensuring sister chromatid cohesion (21, 28, 29). This may be due to its selective loading of PCNA linked to EcoI acetyltransferase (3), which acetylates Smc3, preventing cohesin destabilization by Wapl (30–32). Recruitment of CTF18-RFC to the leading strand is also important for activation of the replication checkpoint (22, 24, 33, 34).

The CTF18-RFC complex features two distinct modules with separate functions (23, 28, 35, 36): a catalytic RFC module for PCNA loading, consisting of a large RFC1-like subunit (hereby referred to as Ctf18), alongside the four RFC2, RFC3, RFC4 and RFC5 subunits utilized by the canonical RFC complex, and a regulatory module, made up of the Ctf8 and Dcc1 subunits. These latter two subunits attach to the Ctf18 C-terminus, forming the Ctf18-1-8 module. A long linker predicted to be flexible connects these two segments (34, 37). Structural and functional studies have demonstrated that the interaction between the Ctf18-1-8 module and the catalytic domain of Pol2 (Pol2_{CAT}), the principal subunit of Pol ϵ , directs CTF18-RFC to replication forks and boosts its clamp loading efficiency (34, 37). This interaction situates the clamp loader close to the primer/template junction, facilitating PCNA loading (34). Pol2_{CAT} is tethered to the remainder of Pol ϵ via an unstructured linker, making it an integral component of the CMGE (Cdc45-Mcm-GINS-Pol ϵ) core replisome complex (38). It has been proposed that Pol ϵ utilizes both the CMG helicase and PCNA as processivity factors to facilitate normal replication rates (39, 40). Tethering by the CMG complex might allow Pol ϵ to dissociate from the 3' end of the

leading strand but stay at the replication fork until leading strand synthesis restarts. Given the relatively weak interaction of Pol ϵ with PCNA (41), it is possible that multiple PCNA loading events on the leading strand via CTF18-RFC are necessary to achieve full Pol ϵ processivity.

Although the structure of the Ctf18-1-8 module in association with Pol2_{CAT} has been elucidated (34, 37), the architecture of the CTF18-RFC module and its interaction with PCNA remain uncharacterized. To investigate the clamp loading mechanism employed by CTF18-RFC, we reconstituted the full human CTF18-RFC complex with PCNA and determined its structure using cryo-electron microscopy (cryo-EM). Our cryo-EM data supports the prediction that the Ctf18-1-8 and RFC modules are flexibly tethered. The analysis yielded high-resolution reconstructions of the entire CTF18-RFC module bound to a closed PCNA ring. Although the overall architecture mirrors the autoinhibited state observed in the canonical RFC loader in a complex with PCNA (10), distinctive features within CTF18-RFC were identified. These distinctions may offer insights into the specialized functional roles CTF18-RFC plays in DNA replication.

Results

Structure of the human CTF18-RFC–PCNA complex

We employed single-particle cryo-EM to determine the structure of human CTF18-RFC bound to PCNA. To this purpose, the full CTF18-RFC complex, including Ctf8 and Dcc1, was expressed in insect cells and purified to homogeneity (Fig S1). The purified complex includes all subunits as shown in the SDS-PAGE gel (Fig S1). With the aim of trapping CTF18-RFC in different steps of the clamp loading reaction, the complex was vitrified in the presence of ATP, PCNA, and primer/template DNA but omitting the Mg²⁺ cofactor to halt hydrolysis (42), and then imaged by cryo-EM (Fig S2). Surprisingly, image processing yielded only a major 3D class, including the CTF18-RFC pentamer bound to a closed PCNA ring (Fig 1a–b, Fig S2-3), with no evidence of DNA density. The Ctf18-1-8 module, which is connected to the RFC module by a 91 amino acid linker (Fig 1a), is not resolved in the map (Fig 1b), suggesting it is flexibly tethered to the RFC module. The map was refined to a global resolution of 2.9 Å (Fig S2-3), which allowed model building of the five CTF18-RFC subunits as well as the PCNA homotrimer (Fig 1c). The final model refines to a map-to-model FSC of 3.0 Å at 0.5 threshold, with good refinement statistics (S1 Table).

The five CTF18-RFC subunits are positioned on top of the closed PCNA ring, which is planar and undistorted (Fig 1c). A structural comparison shows that the CTF18-RFC pentamer was trapped in an autoinhibited conformation analogous to that reported for the canonical human (h) and yeast (sc) RFC (8, 10), pertaining to the initial step of the clamp loading reaction, right before clamp opening (Fig 1d). In such conformation, the AAA+ domains form an overtwisted and asymmetric spiral that is incompatible with DNA binding (8, 10). The AAA+, collar, and A' domains of the Ctf18 subunit (Fig 2a) align with the homologous domains of hRFC1 (Fig 2b). As in the hRFC structure (10), the Ctf18 subunit interacts with RFC2, RFC3, and PCNA-I. Three main deviations from the hRFC–PCNA structure are observed (10): a different tilt of the PCNA ring (Fig 1c–d), significant mobility of the large Ctf18 subunit (Fig 1b and Fig S2) and, importantly, the presence of a β -hairpin engaged to the outer surface of the RFC5 subunit, which we mapped to the disordered N-terminus of the Ctf18 subunit (Fig 2a and 2c). These features are analyzed in detail below.

Structure of the large subunit of CTF18-RFC

The Ctf18 subunit presents 26% identity with the homologous subunit RFC1 of hRFC and an overall structural conservation, with some deviations particularly in the A' domain (Fig 2b). Compared to the other subunits, the AAA+ domain of Ctf18 shows increased flexibility, as evidenced by the local resolution map of the complex as well as 3D variability analysis (43, 44) (Fig S2d and

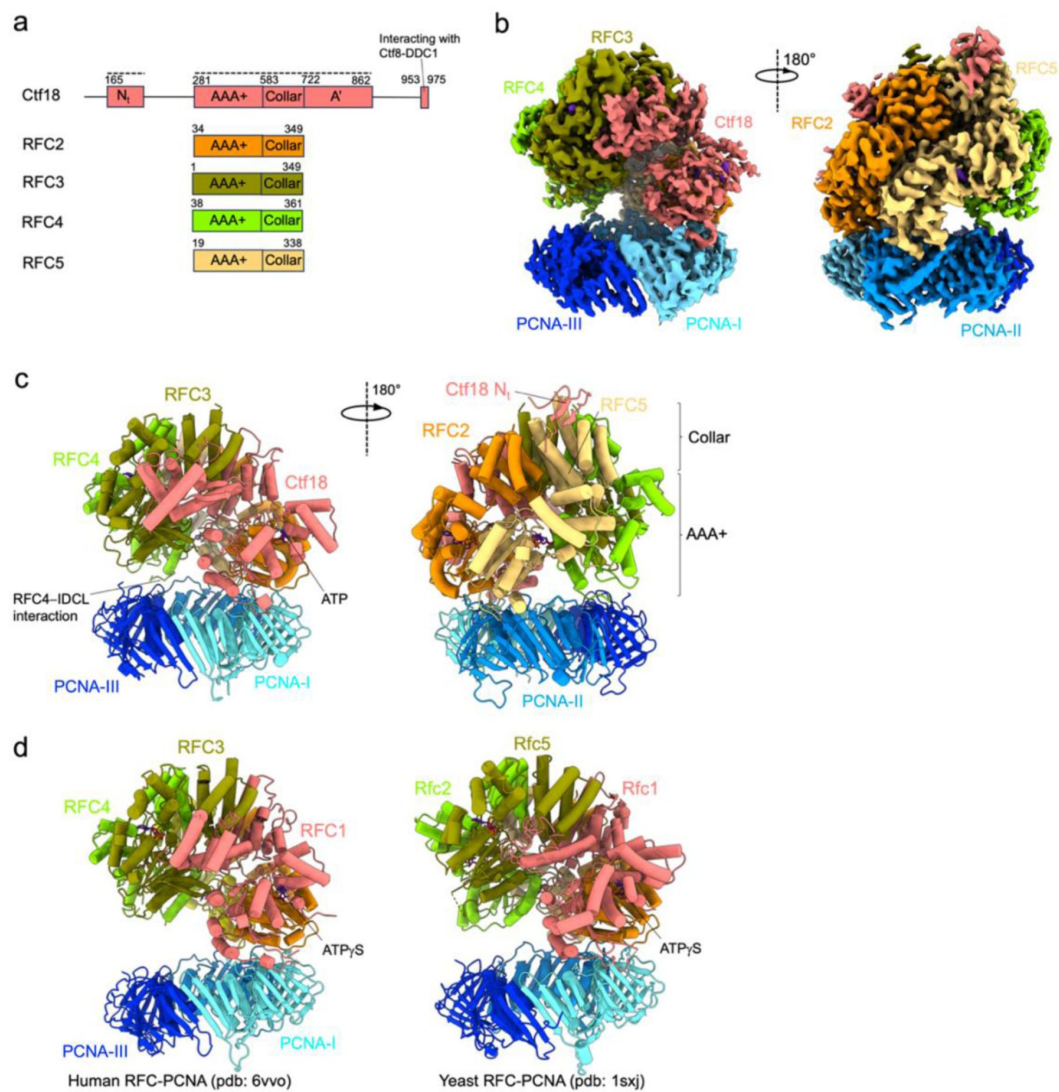


Figure 1.

Cryo-EM structure of the human CTF18-RFC-PCNA complex in the presence of ATP.

a) Domain organization of the CTF18-RFC subunits forming the RFC pentamer. The dotted line above the Ctf18 subunit highlights the regions that are observed in the cryo-EM map. **b)** Two views of the cryo-EM map of the CTF18-RFC-PCNA complex, coloured by subunits. The less sharp definition of the map region corresponding to the AAA+ domain of the Ctf18 subunit supports a relative high mobility of this subunit. **c)** Molecular model of the CTF18-RFC-PCNA complex in ribbon representation. **d)** Structures of the human (left) and *Saccharomyces cerevisiae* (right) RFC-PCNA complex, which present the RFC pentamer in an auto-inhibited conformation analogous to the one captured in the CTF18-RFC-PCNA complex.

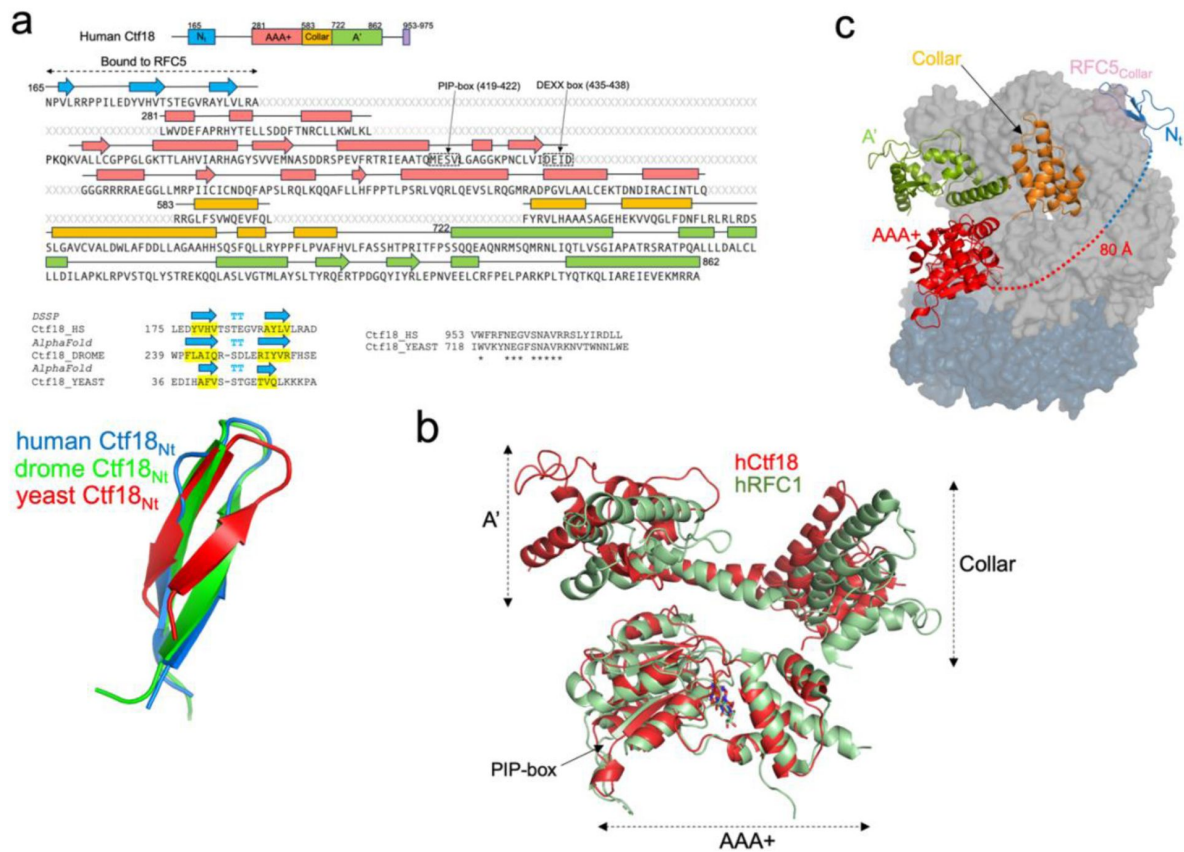


Figure 2.

Structure of the Ctf18 subunit.

a) Sequence of the human Ctf18 subunit, highlighting the secondary structural elements as assigned in the cryo-EM model. The atypical PCNA-interacting motif (PIP-box) and conserved DEX box are boxed. Residues that are not modelled in the cryo-EM structure are labelled as X in grey. Below the Ctf18 sequence to the left, the sequences of the N-terminus of human and other lower eukaryotes Ctf18, together with the secondary prediction by AlphaFold (45) are shown, highlighting the conservation of the β -hairpin fold. Below, the cryo-EM β -hairpin fold model is overlaid onto the corresponding motifs predicted by AlphaFold for *Drosophila* or *Saccharomyces cerevisiae* Ctf18. Below the Ctf18 sequence to the right, the alignment of the C-terminal region of human and *Saccharomyces cerevisiae* Ctf18, which interacts with the Ctf8/Dcc1 subunits, is shown. Asterisks denote conserved residues. **b)** Superposition of the human Ctf18 and RFC1 (PDB: 6VVO) subunit structures, with subdomains labelled. **c)** Structure of the Ctf18 subunit, in ribbon representation, within the CTF18-RFC-PCNA complex. The Ctf18 subdomains are coloured-coded as in panel (a). The other complex subunits are shown as surfaces.

Supplementary Video 1). Increased flexibility was not detected in the RFC1 subunit of human RFC bound to PCNA (10). This discrepancy could be explained by diminished inter-subunit interactions involving both the AAA+ and collar domains of Ctf18 compared to RFC1, as illustrated in Fig S4: the AAA+ domain of Ctf18 forms fewer stabilizing salt bridges and h-bonds with RFC2 than RFC1 does, and the Ctf18 collar domain establishes markedly weaker interactions with both RFC2 and RFC3 than the extensive network observed for RFC1. Conversely, the interaction between the A' domain of CTF18-RFC and RFC3 is strong as observed for RFC. Overall, the total buried surface area between Ctf18 and interacting subunits is comparable to that of RFC1 (~5500 Å² versus ~5300 Å²). However, the buried surface area between Ctf18 and RFC2 is markedly smaller than that of RFC1 (~1400 Å² and ~2300 Å²). As explained below, the lost interactions mediated by the AAA+ and collar domains of Ctf18 are partially counterbalanced by those established with the N-terminal β-hairpin, which is absent in the canonical RFC. Additionally, the interaction between Ctf18 and PCNA is weaker compared to the interaction between RFC1 and PCNA, further contributing to these structural differences (see next section). The Ctf18 C-terminal region (residues 953-975) predicted to interact with the Ctf8 and Dcc1 subunits (Fig 2a) is invisible in the map.

The most striking feature is a small β-hairpin-containing fold mapped between residues 165-194 in the disordered N-terminus (Fig 2a and 2c) of the Ctf18 subunit, ~80 Å apart from the Ctf18 AAA+ domain, to which it is connected by a disordered linker of 86 amino acids (residues 195-280). This N-terminal fold plugs into a groove at the outer surface of the collar domain of the RFC5 subunit defined by helices α1, α3, α4, and α5 (Fig 3a). The resolution of the map in this region is sufficient for confident model building of the interface (Fig 3b). The interface buries 984 Å² and is characterized by both apolar and polar interactions (Fig 3a). A set of hydrophobic side chains protruding from Ctf18 β-hairpin (V178, V180, L190, L192) approaches a complementary patch of hydrophobic residues on RFC5 (L255 in α1; F287, V291 and F293 in α3; I336 and V337 in α5). The interface is further strengthened by several hydrogen bonds mostly mediated by main-chain atoms in the Ctf18 β-hairpin, and by a salt bridge between R193 of the Ctf18 β-hairpin and D292 of RFC5. It is worth highlighting that RFC2 separates the Ctf18 and RFC5 subunits. Despite this separation, Ctf18 projects the β-hairpin directly onto RFC5, bypassing any interactions with RFC2 (Fig 2c). This observation supports that the β-hairpin functions as a “latch” to stabilize the CTF18-RFC structure.

Similarly to human CTF18-RFC, long regions predicted to be disordered by AlphaFold (45) are present in the N-terminal sequence of Ctf18 of lower eukaryotes. The β-hairpin region of human Ctf18 does not show strict sequence conservation, but a similar fold in the N-terminus of lower eukaryotes Ctf18 is predicted (Fig 2a). Therefore, lower eukaryotes may use a similar β-hairpin motif to bind the corresponding subunit of the RFC-module complex (RFC5 in human, Rfc3 in *S. cerevisiae*), emphasizing its importance.

Interactions of CTF18-RFC with PCNA

CTF18-RFC engages two of the three PCNA protomers through four contact points involving the AAA+ domains of Ctf18, RFC2, RFC4, and RFC5 (Fig 1c and Fig 3c). The interactions of PCNA with the RFC2 and RFC5 subunits are analogous to those reported in the hRFC-PCNA complex (10). However, while Ctf18 engages PCNA-I via a PIP-box interaction similar to hRFC1 (10), its atypical PIP motif — lacking aromatic residues (Fig. 2a and Fig. 3c) — results in a smaller buried surface area (~780 Å² vs. ~1010 Å²). A methionine residue (M419) inserts into the so-called “Q-pocket,” which is typically occupied by a glutamine in canonical PIP-boxes, while V422 occupies the conserved hydrophobic pocket that usually accommodates aromatic residues. This low-affinity interaction does not stably anchor Ctf18 to PCNA-I and may contribute to the relative flexibility of the AAA+ domain of Ctf18 observed in the complex (Fig S2 and Supplementary Video 1). The PIP-box of RFC5 is also atypical but contains two hydrophobic residues (I114 and F115) that insert into

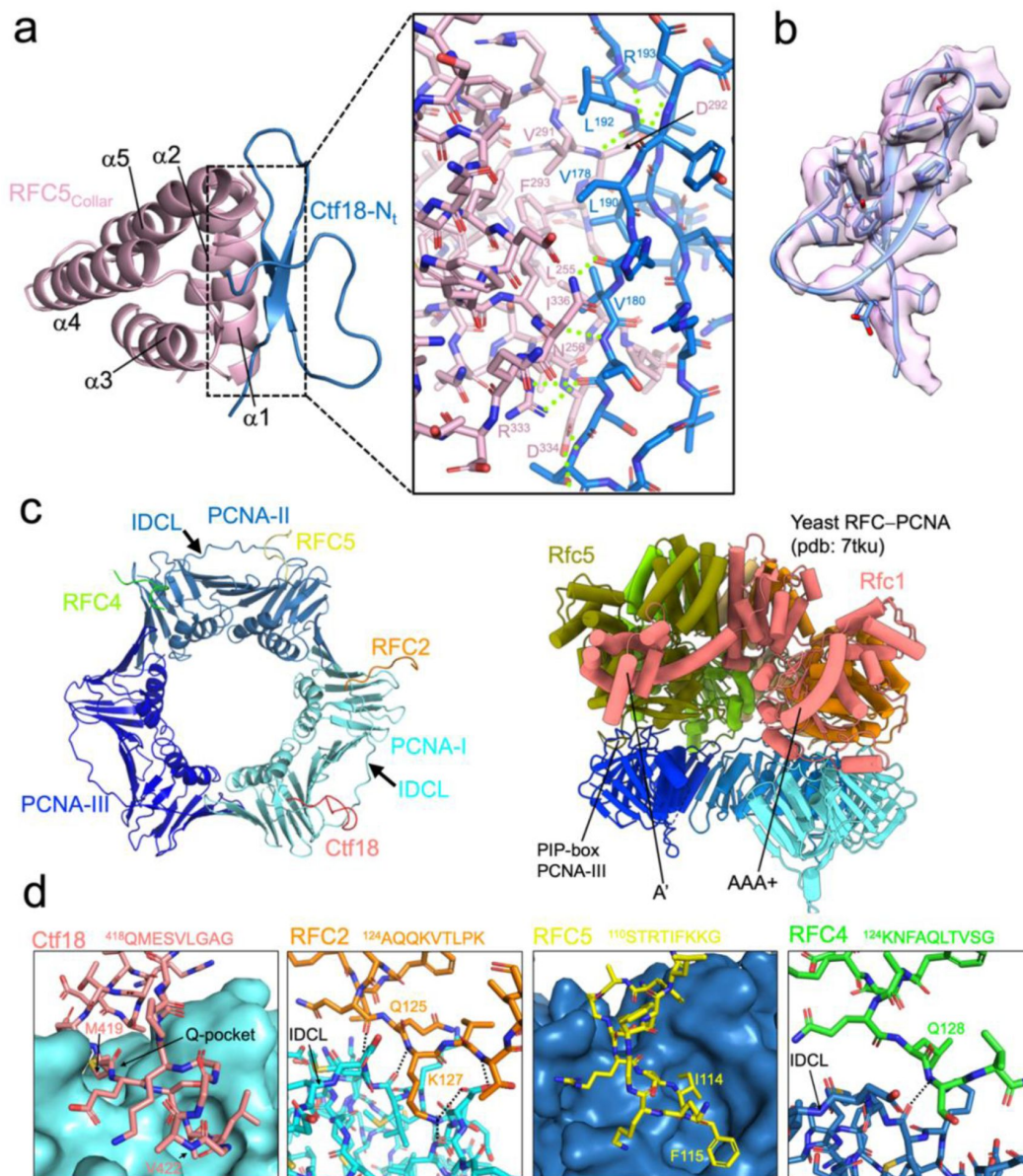


Figure 3.

Structural details of the Ctf18-RFC5 and CTF18-RFC-PCNA interactions.

a) Model of the RFC5 collar domain and Ctf18 N-terminal β -hairpin in ribbon representation. The inset shows the interface in stick representation, with hydrogen bonds shown as green dotted lines. **b)** Map region around the Ctf18 N-terminal β -hairpin. **c)** Model of the PCNA homotrimer bound to the four CTF18 interacting regions. To the right, the structure of the yeast RFC-PCNA complex with an open clamp, highlighting the engagement of all five RFC subunits to PCNA, required for clamp opening. **d)** Model of the regions of the CTF18-RFC-PCNA interactions, with CTF18-RFC subunit residues shown as sticks and PCNA as surface or ribbon. Polar interactions are shown as black dotted lines. The amino acid sequence encompassing the interacting motifs in the various subunits are shown.

the conserved pocket of PCNA-II. The interacting motifs of RFC2 and RFC4 differ from each other in sequence but both engage similar regions involving the interdomain connecting loop (IDCL) of PCNA-I and PCNA-II, respectively (**Fig 3d** [↗](#)).

Interestingly, while RFC4 interacts with PCNA in the CTF18-RFC complex, it does not in the hRFC-PCNA complex ([10](#) [↗](#)). The RFC4-PCNA interaction, which was also observed in the structure of *S. cerevisiae* RFC bound to open PCNA ([11](#) [↗](#)), brings PCNA closer to the CTF18-RFC module, thereby explaining the distinct tilt of the PCNA ring observed in our structure compared to the hRFC-PCNA complex (**Fig. 1c** [↗](#) and **1d** [↗](#)). Given the structure of scRFC bound to open PCNA ([11](#) [↗](#)), where all five RFC subunits are bound to PCNA (**Fig 3d** [↗](#)), we predict that a conformational shift in CTF18-RFC, specifically bringing the fifth subunit (RFC3) to engage PCNA-III, would be necessary to disrupt the interface between PCNA-I and PCNA-III, thereby opening the clamp and aligning the CTF18-RFC subunits for primer-template DNA interaction. The reason for the absence of such an active CTF18-RFC conformation in our cryo-EM analysis remains uncertain. However, similar to observations with the hRFC-PCNA complex, where an open-clamp state was not reported ([10](#) [↗](#)), this could be attributed to the hydrophobic regions of the open clamp interacting with the water-air interface, potentially leading to protein denaturation during the vitrification process to prepare the cryo-EM grids.

Nucleotide binding to the AAA+ modules

The map is consistent with nucleotides in four of the five AAA+ domains of the CTF18-RFC subunits (**Fig 4a** [↗](#)). ATP is bound to subunits Ctf18, RFC2, and RFC5, while ADP appears to engage RFC3, consistent with the structure of the hRFC-PCNA complex reconstituted with the slowly hydrolysable nucleotide ATPγS ([10](#) [↗](#)). Density for the adenine ring of ATP in the Ctf18 subunit is poor, likely a consequence of the increased mobility of the AAA+ domain of this subunit (**Fig 4a** [↗](#)). Compared to the canonical RFC1, the Walker

A motif of Ctf18 contains a leucine residue (L378) instead of a valine, but this substitution does not appear to affect its interaction with the ATP nucleotide (**Fig S5**). Since RFC3 lacks critical catalytic residues and bears a substitution in the Walker A motif ([10](#) [↗](#)), the presence of ADP, rather than a product of ATP hydrolysis, probably results from co-purification of the complex where ADP contaminants are present. In the human RFC complex, it has been previously shown that an intact nucleotide binding site in RFC3 has an important structural role in the complex assembly ([46](#) [↗](#), [47](#) [↗](#)). If this is the case for CTF18-RFC, it remains to be established, but it is likely that ADP binding to this subunit may play a stabilizing role in all RFC-like loaders.

Differently from the hRFC-PCNA complex, where RFC4 is bound to ATPγS, in CTF18-RFC, this subunit is not engaged with a nucleotide (**Fig 4a** [↗](#)). The residues in the AAA+ of RFC4 of CTF18-RFC coordinated to ATPγS in hRFC are displaced and incompatible with nucleotide binding (**Fig 4c** [↗](#)). In particular, the side chains of R45 and P46 in RFC4 occupy the space where ATP would normally bind, thereby sterically hindering nucleotide accommodation in the CTF18-RFC structure (**Fig 4c** [↗](#)). Interestingly, mutation of the conserved K84 residue in the ATP-binding motif of RFC4 in hRFC did not impair the ability of this subunit to assemble with other RFC subunits ([46](#) [↗](#), [47](#) [↗](#)). In agreement, our structure shows that the lack of ATP binding to RFC4 does not preclude the assembly of the CTF18-RFC pentamer in the autoinhibited conformation. However, considering the importance of ATP binding of RFC4 for the hRFC clamp loading function ([48](#) [↗](#)–[51](#) [↗](#)), it is expected that ATP binding to RFC4 in CTF18-RFC is required to switch to an enzymatically active conformation.

The asymmetric spiral of the CTF18-RFC AAA+ pentamer (**Fig 4a-b** [↗](#)) disrupts the active sites at the Ctf18/RFC2, RFC2/5, and RFC5/4 interfaces (**Fig 4c** [↗](#)). At these interfaces, the *trans*-acting arginine fingers are positioned too far away to interact with the γ-phosphate of ATP (e.g., ~9 Å for R184 of RFC2, ~10 Å for R168 of RFC5, and ~7 Å for R193 of RFC4) (**Fig 4c** [↗](#)). Prior research has

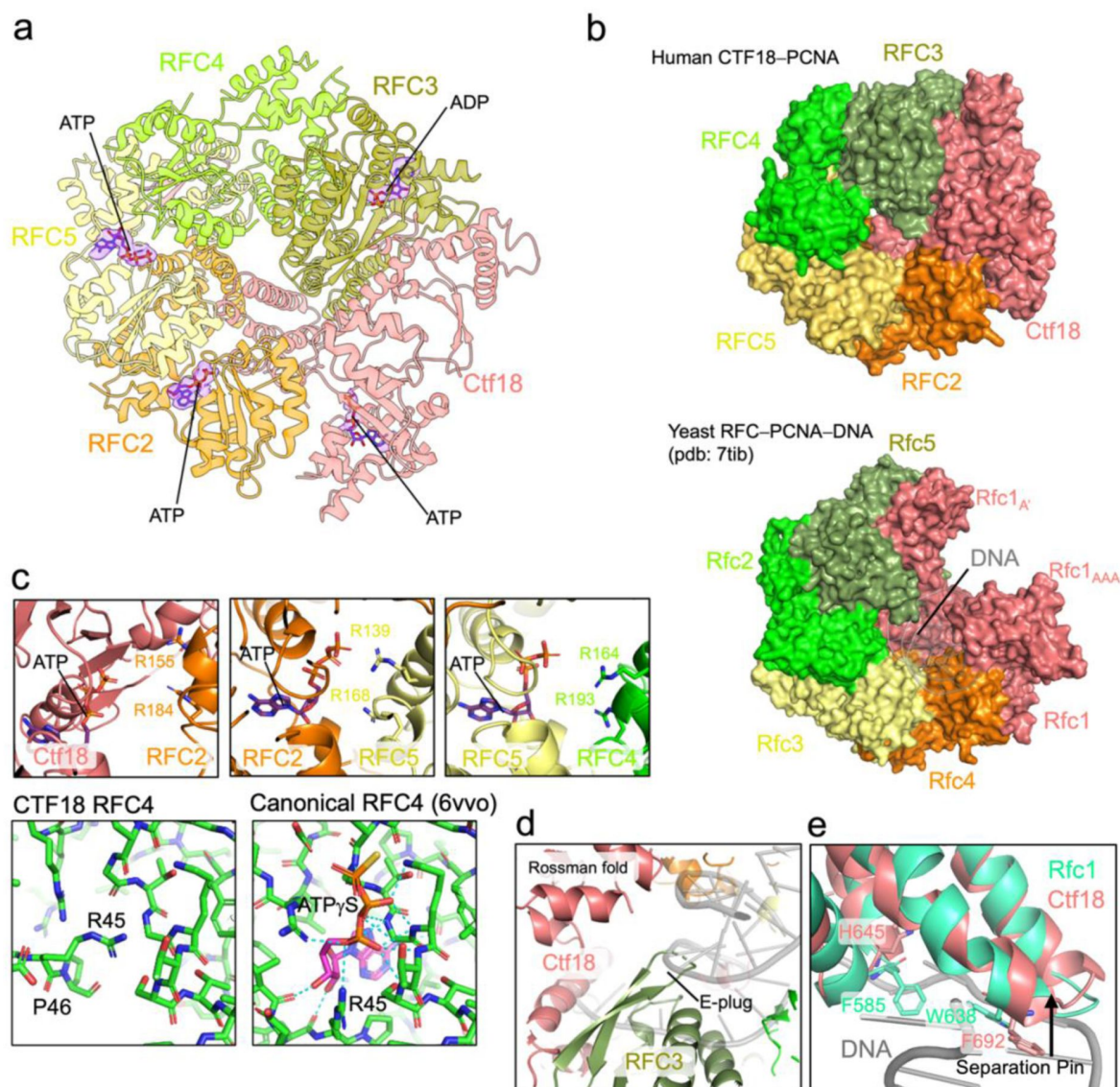


Figure 4.

Analysis of the auto-inhibited conformation of the CTF18 pentamer and nucleotide coordination.

a) Ribbon model of the CTF18-RFC pentamer (PCNA hidden). Nucleotide molecules are shown as sticks, with the corresponding cryo-EM density. **b**) Surface representation of (top) the CTF18-RFC pentamer and (bottom) the yeast RFC pentamer bound to open PCNA and primer-template DNA. PCNA is hidden in both structures. The overtaken CTF18-RFC pentamer cannot accommodate DNA within the inner chamber. **c**) Model of the ATP active sites in the CTF18-RFC-PCNA complex, showing disruption of the interactions mediated by the arginine fingers. Below, insets show the nucleotide-binding site of subunit RFC4 in either the human CTF18-RFC (left) or human RFC (right) models, highlighting the structural incompatibility with nucleotide binding in the CTF18-RFC structure. Polar interactions with ATP_γS in human RFC4 are indicated as cyan dotted lines. **d**) Detail of the inner chamber region of the CTF18-RFC pentamer, showing the β-hairpin (E-plug) of RFC5 protruding in the region predicted to be occupied by DNA. The DNA (grey ribbon) is modeled as in the yeast RFC-DNA-PCNA complex. **e**) The structures of the RFC1 subunit from the yeast RFC-DNA-PCNA complex (11) and the Ctf18 subunit from the CTF18-RFC-PCNA complex are aligned to highlight conservation of the separation pin. Side chains of residues implicated in this motif in yeast RFC, along with the corresponding amino acids in CTF18-RFC, are shown.

indicated that hydrolysis at the large subunit/RFC2 interface is not essential for clamp loading by various loaders (48–51), while the others are critical for the clamp-loading activity of eukaryotic RFCs.

In CTF18-RFC, the AAA⁺ spiral is overtwisted in a manner reminiscent of the autoinhibited conformations observed in hRFC and scRFC (8, 10) (**Fig 1d**). Specifically, the overtwisted RFC3 and RFC4 subunits occupy the space normally filled by DNA. In the autoinhibited structure of hRFC, the β -hairpin (residues 68–88) located at the base of RFC3 also referred to as “E-plug”, extends towards the anticipated location of DNA and interacts with the Rossmann fold of RFC1 (10). Conversely, in CTF18-RFC, while this same β -hairpin is oriented toward the interior of the DNA-binding channel (**Fig 4d**), it does not make direct contact with the Ctf18 subunit. Collectively, the E-plug appears a conserved feature in eukaryotic clamp loaders, underlying its importance in maintaining the pentamer in its autoinhibited conformation, by preventing DNA access through the central channel. We observed that human CTF18-RFC shares an additional structural feature with both human and yeast RFC: the presence of a separation pin in the large subunit (10, 11). Although the separation pin is not essential for clamp loading (11), its conservation across species and clamp loader types suggests an auxiliary or context-dependent function. Interestingly, residues W638 and F582—proposed to stabilize the separation pin and its interaction with DNA in yeast RFC—are substituted by F692 and H645 in human CTF18-RFC, respectively (**Fig 4e**), possibly indicating a mechanistically similar role in both systems.

To discount the possibility that the absence of the Mg^{2+} ions in the cryo-EM sample could have hindered the proper assembly of the CTF18-RFC pentamer for DNA engagement, we used cryo-EM to visualize CTF18-RFC with PCNA, primer-template DNA, ATP, and Mg^{2+} (**Fig S6–7**, S1 Table). Despite the inclusion of Mg^{2+} , the resulting 3.2 Å reconstruction of the complex remained in a configuration analogous to that without Mg^{2+} (**Fig 5a–b**, **Fig S6**). In this reconstruction, distinct density for Mg^{2+} is noted in the active sites of RFC2 and RFC5, coordinating ATP (**Fig 5b**). Similar to the map without Mg^{2+} , ADP coordination is present in RFC3, while RFC4 is not engaged with any nucleotide. These findings collectively imply that the presence of Mg^{2+} in the nucleotide binding sites does not influence the transition of CTF18-RFC from an autoinhibited to an active state. This result also confirms the remarkable stability of the autoinhibited conformation, even in hydrolysing conditions.

The Ctf18 β -hairpin plays a role in PCNA loading and in stimulating DNA synthesis by Pol ϵ

Based on the structures reported herein, the β -hairpin at the disordered N-terminal domain of Ctf18 is a distinctive feature that may serve to stabilize the CTF18-RFC pentamer and maximize its clamp loading efficiency. To investigate the functional importance of the β -hairpin, we purified a CTF18-RFC mutant with a truncation at the disordered N-terminal domain of the Ctf18 subunit, specifically, the segment spanning residues 165–194 (CTF18 ^{Δ 165–194}-RFC). This mutation eliminates the entire Ctf18 β -hairpin and is expected to destabilize the CTF18-RFC pentamer. In agreement, when mutant CTF18-RFC was imaged by cryo-EM with PCNA in the same conditions used for wild type CTF18-RFC, only a small subset of particles of the complex was observed, while the majority of particles contained dissociated PCNA (**Fig S8–9**). We hypothesized that this destabilizing mutation in CTF18-RFC may reduce its efficiency in loading PCNA onto DNA.

To test this, and to directly compare CTF18-RFC activity to the canonical clamp loader RFC, we used a pre-steady state ensemble FRET assay to monitor loading of Cy5-labeled PCNA onto a Cy3-labeled primer-template DNA substrate pre-coated with RPA (**Fig 6** and **Fig S10**). Upon addition of RFC, CTF18-RFC, or the mutant CTF18 ^{Δ 165–194}-RFC complex, a time-resolved increase in FRET signal was observed, corresponding to productive PCNA loading. While both CTF18 and CTF18 ^{Δ 165–}

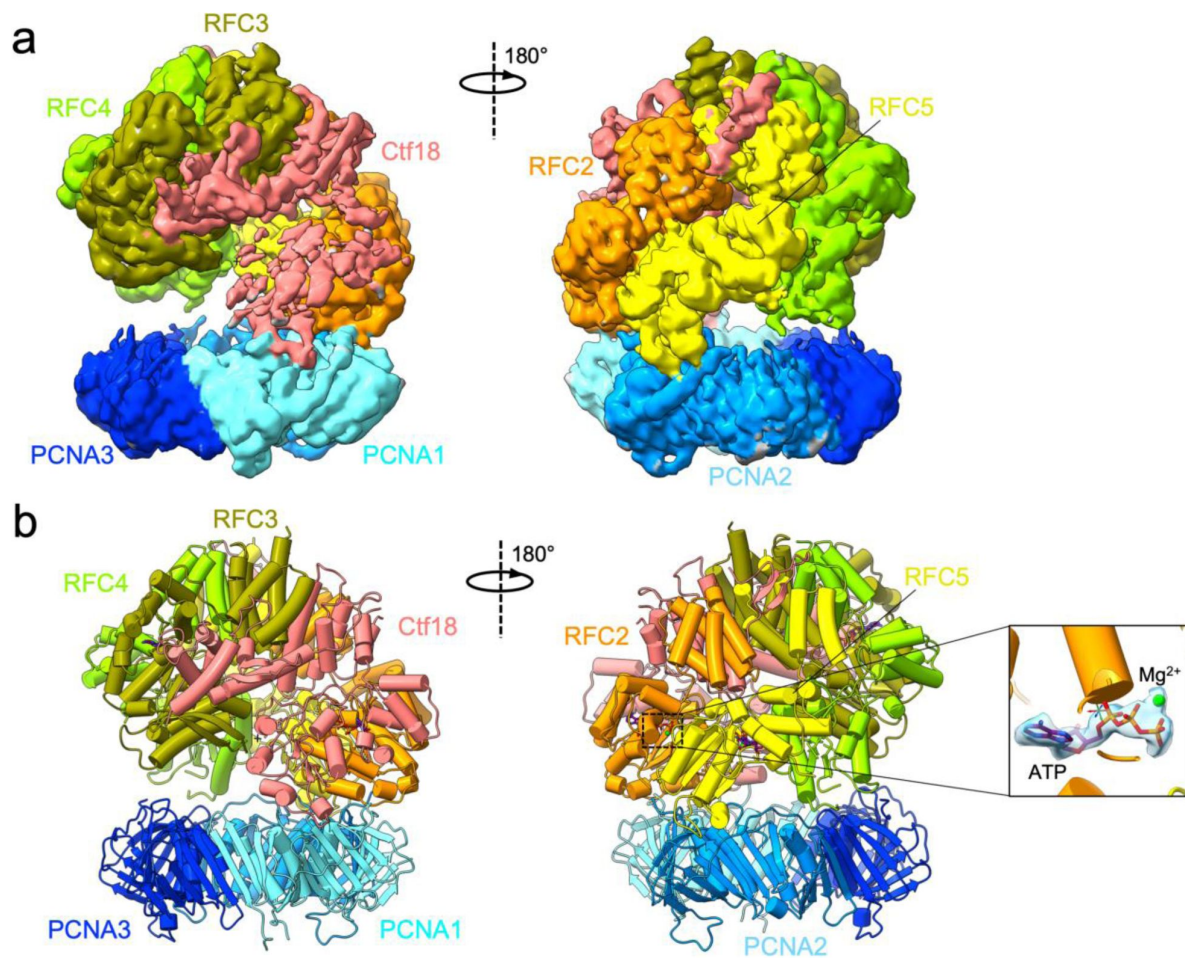


Figure 5.

Cryo-EM structure of the human CTF18-RFC-PCNA complex in the presence of ATP and Mg^{2+} .

a) Two views of the Cryo-EM map, coloured by subunits. **b)** Molecular model in ribbon representation. The inset shows a map and model of the ATP molecule bound to the RFC2 subunit, with a discernible Mg^{2+} ion.

¹⁹⁴-RFC were able to load PCNA onto DNA, the loading kinetics differed significantly: CTF18-RFC displayed a slightly faster rate than RFC, whereas CTF18^{Δ165-194}-RFC exhibited a >2-fold reduction in the fast-phase rate constant (k_1), consistent with a functional defect (**Table 1**). Importantly, the FRET amplitude was comparable between the two CTF18-RFC isoforms, indicating that the mutation does not prevent PCNA loading per se but affects the rate at which loading occurs. These results suggest that the β -hairpin contributes to the efficiency—but not the capacity—of PCNA loading by CTF18-RFC.

We then tested whether the observed loading defect observed with CTF18^{Δ165-194}-RFC results in diminished stimulation of primer synthesis by Pol ϵ . To test this hypothesis, we performed primer extension assays using Pol ϵ , PCNA, and either wild-type or mutant CTF18-RFC constructs to assess their ability to stimulate DNA synthesis by Pol ϵ . Our results indicate a significant reduction in primer extension products when the β -hairpin was deleted (**Fig 7a**). Measurement of band intensities revealed an approximately two-fold decrease in the amount of extended primer produced by the mutant compared to the wild-type (**Fig 7b**). To further support that the observed effect on primer extension is due to the mutation, we conducted a concentration titration comparing wild-type and mutant CTF18-RFC (Fig S11). The CTF18-RFC mutant displayed altered concentration dependencies compared to the wild-type, thereby validating the mutant phenotype and underscoring the structural importance of the β -hairpin feature. Notably, the primer extension assay was conducted with 2 nM plasmid DNA and excess CTF18-RFC (40 nM), PCNA (100 nM), and Pol ϵ (30 nM), establishing single-turnover conditions per DNA molecule. Under these conditions, the rate—not the extent—of PCNA loading becomes the key determinant of synthesis dynamics. Although Pol ϵ can initiate synthesis independently of PCNA, the clamp is required for full processivity. Therefore, the >2-fold slower loading rate observed for the CTF18-RFC mutant likely delays formation of highly processive complexes, resulting in a slower accumulation of extended products and explaining the observed ~2-fold reduction in synthesis.

Discussion

In this work, we reported the structure of the human CTF18-RFC clamp loader bound to PCNA in the presence of ATP with or without Mg^{2+} ions. Our cryo-EM data, together with previous work (34, 37), support that the Ctf8 and Dcc1 subunits of CTF18-RFC, forming the regulatory module interacting with Pol ϵ (37), are flexibly tethered to the RFC-module via binding to the Ctf18 C-terminus, which is separated by the N-terminal RFC1-like module of Ctf18 by ~90 amino acid flexible linker. Therefore, CTF18-RFC comprises two structurally distinct modules that cooperate to load PCNA onto the leading strand, facilitating replication by Pol ϵ .

The CTF18-RFC-PCNA complex was observed in an autoinhibited conformation, which prevents DNA binding. This mirrors the similar conformation in hRFC and scRFC, as previously reported (10, 11). Such conformation represents a transient encounter complex that occurs early in the clamp loading process, before the opening of the clamp. Despite the variations in the large RFC subunit in the two assemblies, we demonstrate a striking architectural conservation between the alternative human clamp loader CTF18-RFC and the canonical RFC. Specifically, CTF18-RFC utilizes four subunits (Ctf18, RFC2, RFC5, and RFC4) to engage two PCNA protomers. However, the engagement of the third PCNA protomer by RFC3 is necessary to initiate the opening of the PCNA ring interface, a process that requires a conformational shift of the AAA⁺ pentamer from an autoinhibited to an active state. Supporting this, previous structural studies on scRFC have shown that the conformation of RFC bound to open PCNA is facilitated through a 'crab-claw' mechanism, which is inherently predisposed to DNA binding (11). This conformational change creates an opening between the A' segment and the AAA⁺ module in the A subunit, allowing primer-template DNA to bind within the RFC inner chamber (11) (**Fig 4C**). Remarkably, this significant conformational shift occurs rapidly (52), does not depend on ATP hydrolysis, and ensures the sequential binding of PCNA and the DNA substrate in the correct order. While our study did not

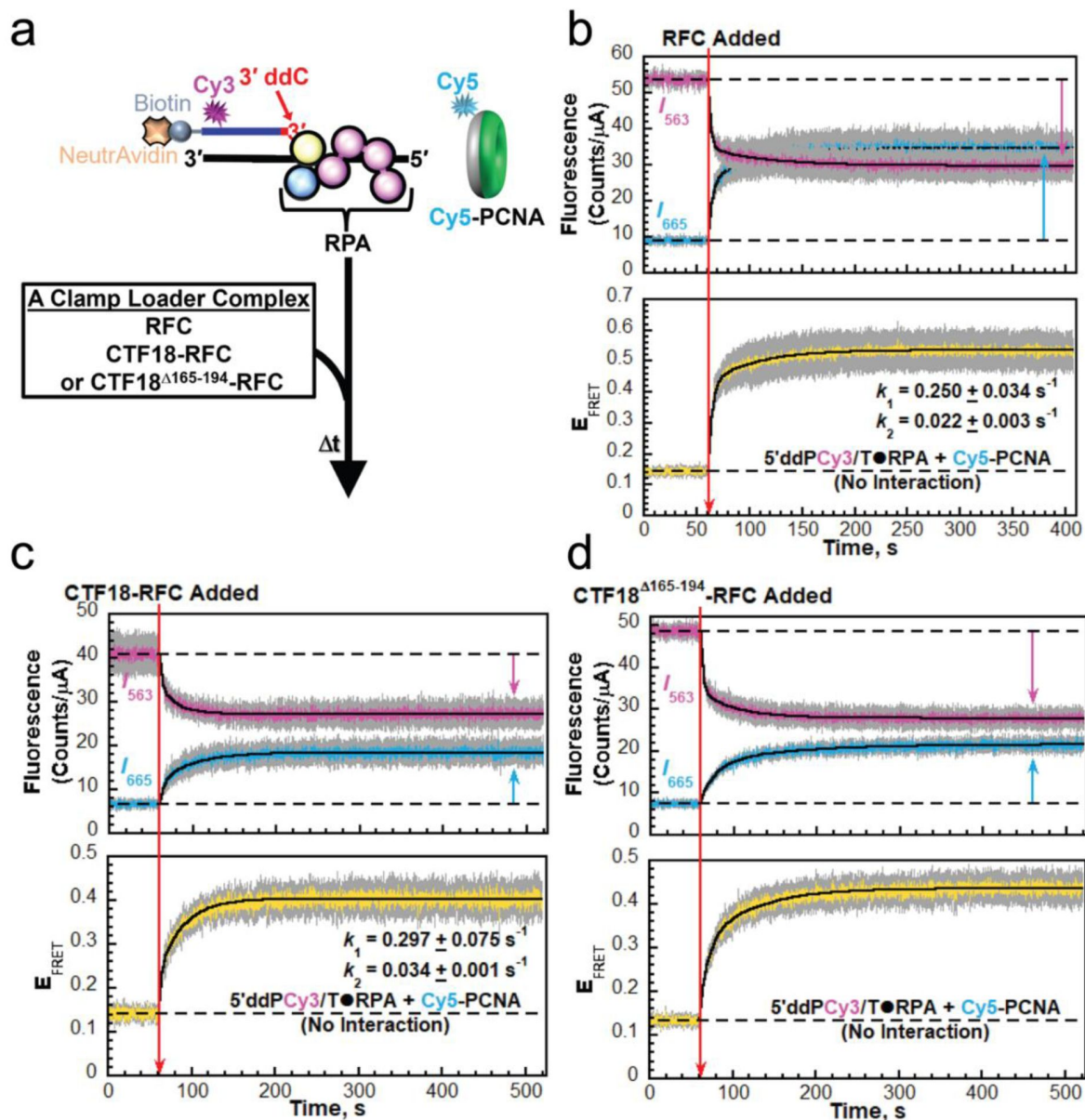


Figure 6.

Loading of PCNA onto P/T junctions by human clamp loader complexes.

a) Schematic representation of the FRET pair and experiment to monitor loading of Cy5-PCNA onto 5'ddPCy3/T DNA substrates engaged by RPA. The front and back faces of PCNA are displayed in green and grey, respectively. When loaded onto a 5'ddPCy3/T DNA substrate, the Cy5 label on the back face of PCNA is oriented towards the Cy3 label near the blunt duplex end of the 5'ddPCy3/T DNA substrate, yielding a FRET signal. **b-d)** Data. Each trace is the mean of at least three independent traces with the S.E.M. shown in grey. The time trajectories of I_{563} (magenta) and I_{665} (cyan) are displayed in the top panels and the corresponding E_{FRET} (mustard) is displayed in the bottom panels. The time at which a clamp loader complex is added is indicated by a red arrow. Changes in I_{563} and I_{665} are indicated in the top panel by magenta and cyan arrows, respectively. The I_{563} , I_{665} , and the corresponding E_{FRET} values observed prior to the addition of clamp loader complexes represents the complete absence of interactions between 5'ddPCy3/T-RPA complexes and Cy5-PCNA. For observations, these values are each fit to flat line that is extrapolated to the axis limits. For observation, the I_{563} and I_{665} values observed after the addition of clamp loader complexes are fit to double exponential declines (I_{563}) and rises (I_{665}), respectively. The E_{FRET} values observed after the addition of a given clamp loader are fit to double exponential rises and the respective rate constants (k_1 and k_2) are reported in the corresponding figure and in Table 1. Data for loading of PCNA by RFC, CTF18-RFC, and CTF18 $^{\Delta 165-194}$ -RFC are displayed in panels **b**, **c**, and **d**, respectively.

Table 1.

Kinetic variables for loading of PCNA by human clamp loader complexes.

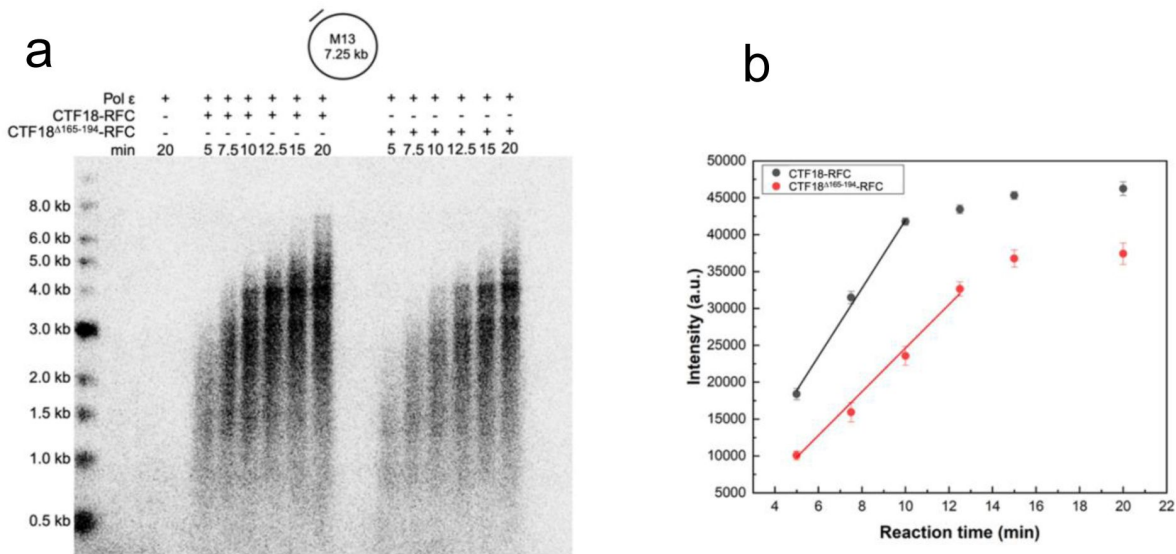
Kinetic variables are from the fits to double exponential rises for the data presented in Figure 6 in the main text. The values in parentheses indicate the relationship (as a fraction) between the rate constants observed for CTF18-RFC and CTF18 Δ 165-194-RFC.

Clamp Loader	Kinetic Variables			
	A_1/A_T	k_1, s^{-1}	A_2/A_T	k_2, s^{-1}
RFC	0.728 ± 0.033	0.250 ± 0.034	0.272 ± 0.033	0.022 ± 0.003
CTF18-RFC	0.356 ± 0.028	0.297 ± 0.075 (1.00)	0.644 ± 0.028	0.034 ± 0.001 (1.00)
CTF18 Δ 165-194-RFC	0.557 ± 0.023	0.108 ± 0.007 (0.36)	0.443 ± 0.023	0.017 ± 0.001 (0.49)

Figure 7.

Primer extension assay with Pol ϵ and CTF18-RFC WT or CTF18 Δ 165-194-RFC.

a) Time course reactions on M13mp18 single-strand DNA with 40 nM wildtype CTF18-RFC or CTF18 Δ 165-194-RFC. Reactions were performed as described in Materials and Methods. **b)** Quantification of the band intensities from the primer extension assay shows that the rate of increase in intensity, derived from the initial linear portion of the reaction time before saturation, is 4142 ± 95 a.u./min for CTF18-RFC and 2389 ± 143 a.u./min for CTF18 Δ 165-194-RFC. This indicates that the mutant slows Pol ϵ synthesis by 42%. The experiment was conducted three times. Dots in **b)** represent mean values from the three replicas.



resolve the active conformation of CTF18-RFC, the observed structural conservation in the autoinhibited state, particularly of the large Ctf18 subunit (**Fig 2**), suggests that a similar 'crab-claw' mechanism may facilitate the transition of CTF18-RFC to its active state for PCNA opening and DNA binding. While our work was being finalized, several cryo-EM structures of human CTF18-RFC bound to PCNA, and primer/template DNA were reported by another group (53). These findings are consistent with the distinct features of CTF18-RFC observed in our structures and independently support the notion of significant mechanistic similarity between CTF18-RFC and canonical RFC in loading PCNA onto a ss/dsDNA junction.

The main structural differences between CTF18-RFC and the canonical RFC include increased mobility of the large subunit and the presence of a β -hairpin that attaches to the RFC5 subunit. The mobility of the large subunit is likely enhanced by the low-affinity PIP motif that anchors it to the PCNA ring. We propose that the β -hairpin may have evolved to counterbalance the mobility of the large subunit, helping to maintain its connection to the RFC pentamer and stabilizing the clamp loader structure. Our structural data and functional assays indeed show that deletion of the β -hairpin results in reduced stability of the CTF18-RFC-PCNA complex, slower clamp loading activity, and diminished activity of CTF18-RFC in stimulating Pol ϵ primer synthesis, likely due to reduced CTF18-RFC efficiency in PCNA loading (**Fig 8**). Interestingly, a recent work reporting cryo-EM structures of the yeast homolog of CTF18-RFC bound to PCNA (54) does not support the presence of a stable β -hairpin at the N-terminus of the Ctf18 subunit engaged to the collar of the Rfc3 subunit. Therefore, the functional β -hairpin we observed in human CTF18-RFC appears a distinct feature that pertains to higher eukaryotes, required to maximize the activity of CTF18-RFC in loading PCNA onto the leading strand.

Materials and Methods

Protein expression and purification

Preparation of CTF18-RFC

MultiBac™ expression system (Geneva Biotech) was used to express human Ctf18-RFC2-5 along with two additional cohesion-specific factors, Dcc1 and Ctf8 (named hereafter CTF18-RFC) in Sf9 insect cells. For this purpose, insect cells optimized sequences of N-terminus twin Strept-tagged TEV Ctf18, RFC2, and RFC4 were cloned in pACEBac1 plasmid with independent promoters and terminators, and N-terminus 6X histidine-tagged RFC3, RFC5, Dcc1 and Ctf8 were cloned in pIDS plasmid with independent promoters and terminators by GenScript. Finally, the single transfer vector with different subunit assemblies was generated using *cre* recombinase according to the MultiBac™ expression system user manual. CTF18-RFC encoding all seven subunits in a single MultiBac™ expression plasmid (pACEBac1) was transformed into DH10MultiBac™ cells to make bacmid DNA. To prepare the baculovirus, bacmid DNA containing all subunits was transfected into Sf9 cells using FuGENE® HD (Promega) according to the manufacturer's instructions. This baculovirus prep was amplified twice to obtain a higher titer virus (P3 virus). The expression of CTF18-RFC then proceeded by transfecting a 6 L Hi5 suspension culture at a density of 2×10^6 cells/mL with the P3 virus for 60-65 hours.

Cells were collected by centrifugation at 5,500 xg for 10 minutes and re-suspended in lysis buffer [50 mM Tris-HCl (pH 7.5), 300 mM NaCl, 20 mM imidazole, 5 mM BME, 0.1% Nonidet P-40, 10% glycerol and one EDTA-free protease inhibitor cocktail tablet/50 mL]. All further steps were performed at 4° C. Cells were sonicated and debris was removed by centrifugation at 95,834 xg for 1 hour at 4° C. The resultant supernatant was directly loaded onto a HisTrap HP 5 mL affinity column (Cytiva) pre-equilibrated with buffer A [50 mM Tris-HCl (pH 7.5), 300 mM NaCl, 20 mM imidazole, 5 mM BME and 10% glycerol]. After loading, the column was washed with 50 mL of

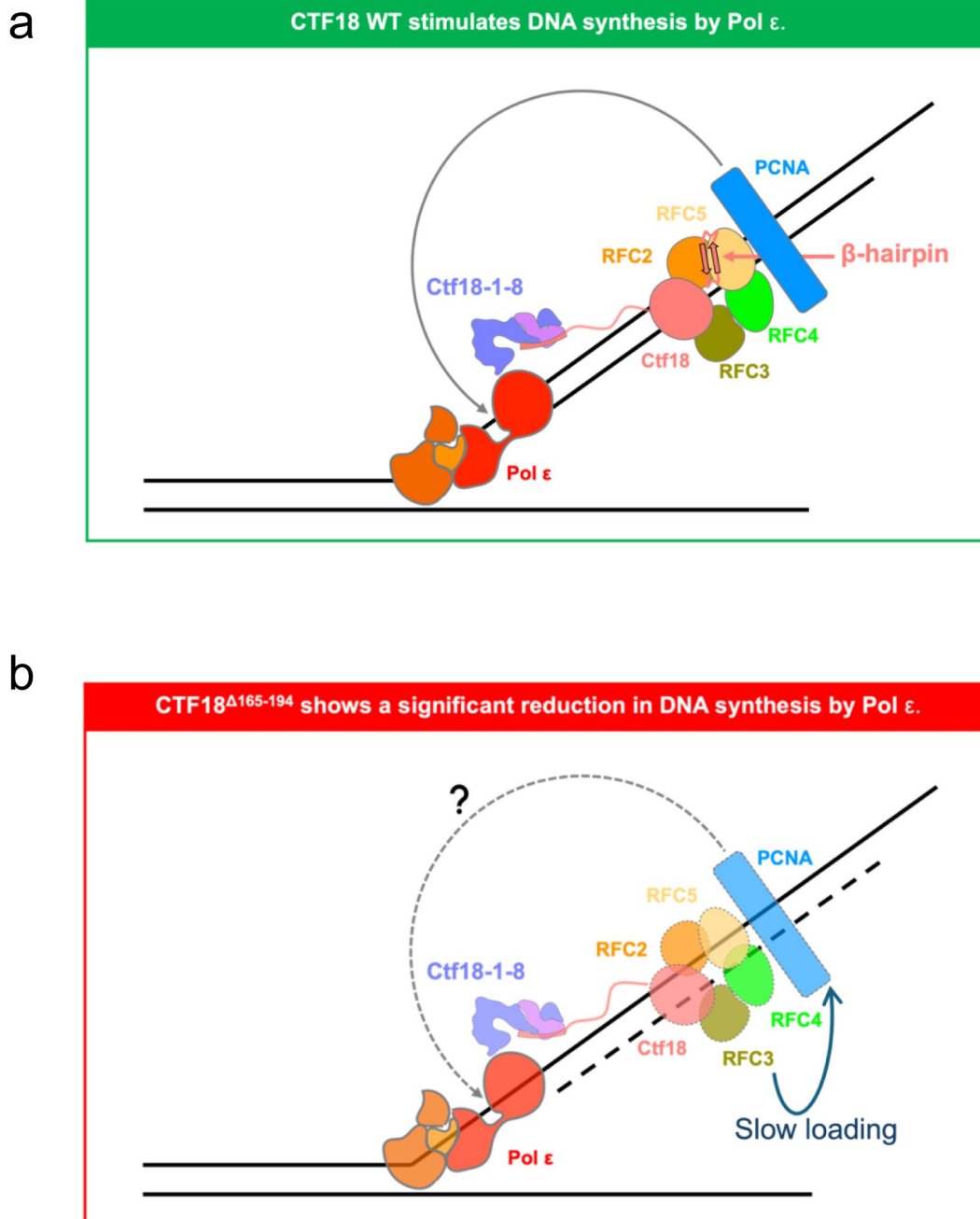


Figure 8.

Deletion of the N-terminal β -hairpin in the Ctf18 large subunit slows down PCNA loading and reduces DNA synthesis by Pol ϵ .

a) The structure of the RFC module of CTF18-RFC is stabilized by the presence of a β -hairpin in the Ctf18 subunit (amino acids 165-194). Thus, wild type CTF18-RFC loads PCNA onto DNA efficiently and stimulates DNA synthesis by Pol ϵ interacting with the polymerase. **b)** The stability of the CTF18-RFC-PCNA complex is diminished by the deletion of the β -hairpin. Consequently, a CTF18-RFC deletion mutant lacking the β -hairpin (CTF18 Δ 165-194-RFC) shows slower loading of PCNA and lower activity in stimulating primer synthesis by Pol ϵ .

buffer A and the bound fractions were eluted by gradient with 50 mL of buffer B [50 mM Tris-HCl (pH 7.5), 300 mM NaCl, 500 mM imidazole, 5 mM BME, and 10% glycerol]. The eluents were pooled and directly loaded onto Strept Trap XT 1 mL affinity column pre-equilibrated with buffer A [50 mM Tris-HCl (pH 7.5), 300 mM NaCl, 5 mM BME and 10% glycerol]. After loading, the column was washed with 10 mL of buffer A, followed by step elution with buffer B [50 mM Tris-HCl (pH 7.5), 300 mM NaCl, 50 mM Biotin, 5 mM BME, and 10% glycerol]. Eluents were pooled and incubated with the TEV protease overnight. Fractions were concentrated to 1 mL and loaded onto HiLoad 16/600 Superdex 200 pg pre-equilibrated with gel filtration buffer [50 mM Tris-HCl (pH 7.5), 200 mM NaCl, 1 mM DTT, and 10% glycerol]. Protein fractions were pooled in a total volume of 1.6 mL, flash frozen, and stored at -80°C.

Preparation of PCNA

To express PCNA BL21(DE3) *E. coli* cells were transformed with WT PCNA plasmid. The transformed cells were grown at 37°C in 2YT media supplemented with ampicillin to an OD₆₀₀ of 1.2. The cell cultures were induced with 0.5 mM isopropyl-beta-D-1-thiogalactopyranoside (IPTG) and let them grow for 19 hours at 16°C. Cells were then harvested by centrifugation at 5500 xg for 10 minutes and then re-suspended in lysis buffer [50 mM Tris-HCl (pH 7.5), 750 mM NaCl, 20 mM imidazole, 5 mM BME, 0.2% Nonidet P-40, 1 mM PMSF, 5% glycerol and one EDTA-free protease inhibitor cocktail tablet/50 mL]. The cells were lysed with 2 mg/mL of lysozyme at 4°C for 30 minutes. Cells were sonicated and debris was removed by centrifugation at 22,040 xg for 20 minutes at 4°C. The cleared cell lysate was directly loaded onto a HisTrap HP 5 mL affinity column (Cytiva) pre-equilibrated with buffer A [50 mM Tris-HCl (pH 7.5), 500 mM NaCl, 20 mM imidazole, 5 mM BME and 5% glycerol]. The column was firstly washed with 50 mL of buffer A and then with 50 mL of buffer B containing low salt [50 mM Tris-HCl (pH 7.5), 100 mM NaCl, 500 mM imidazole, 5 mM BME, and 5% glycerol]. The eluents were pooled and directly loaded onto HiTrap Q HP 5 mL anion exchange column (GE Healthcare) pre-equilibrated with buffer D [50 mM Tris-HCl (pH 7.5), 100 mM NaCl, 5 mM BME and 5% glycerol]. The column was washed with 50 mL of buffer D after loading, followed by elution of 50 mL gradient with buffer E [50 mM Tris-HCl (pH 7.5), 1 M NaCl, 5 mM BME, and 5% glycerol]. Eluents were pooled and concentrated to 1.5 mL and loaded onto HiLoad 16/600 Superdex 200 pg pre-equilibrated with gel filtration buffer [50 mM Tris-HCl (pH 7.5), 150 mM NaCl, 1 mM DTT, and 5% glycerol]. Protein fractions were pooled, flash-frozen, and stored at -80 °C. Human RPA, Cy5-PCNA, and RFC for the clamp loading assay were obtained as previously described ([55](#), [56](#)). The concentration of active RPA was determined via a FRET-based activity assay as described previously ([57](#)).

Preparation of DNA polymerase ε

MultiBac™ expression system (Geneva Biotech) was used to express human DNA Pol ε in Sf9 insect cells. For this purpose, insect cells optimized sequence of the catalytic subunit, p261 was cloned in pACEBac1 and p59, C-terminus TEV twin Strept-tagged p17, and N-terminus 8X histidine-tagged TEV p12 along with independent promoters and terminators were cloned in pIDS by the GenScript. Finally, the single transfer vector with different subunit assemblies was generated using *cre* recombinase according to MultiBac™ expression system user manual. To prepare the baculovirus, the bacmid DNA of Pol ε was transfected into Sf9 cells using FuGENE® HD (Promega) according to the manufacturer's instructions. The resulting supernatant was obtained as the P1 virus stock which was then amplified to obtain P2 virus stock. P2 virus stock was then further amplified to obtain P3 virus stock for large-scale expression. Pol ε was expressed by transfecting 4 L of Sf9 suspension culture at 2×10^6 cells/mL density with P3 virus. Cells were harvested after 72 hours by centrifugation at 5,500 xg for 10 minutes and re-suspended in 200 mL of lysis buffer [50 mM HEPES (pH 8), 20 mM Imidazole, 400 mM NaCl, 5 mM β-Mercaptoethanol, 5% (v/v) Glycerol, 0.1% NP-40, 1 mM PMSF and EDTA-free protease inhibitor cocktail tablet/50 mL (Roche, UK)]. Cells were sonicated and debris was removed by centrifugation at 95,834 xg for 1 hour at 4 °C. The resultant supernatant was directly loaded onto a HisTrap HP 5 mL affinity column (Cytiva) equilibrated with buffer A [50 mM HEPES (pH 8), 20 mM Imidazole, 400 mM NaCl, 5mM β-Mercaptoethanol, 5%

(v/v) Glycerol and EDTA-free protease inhibitor cocktail tablet/50 mL (Roche, UK). After loading, the column was washed with 50 mL of buffer A and the bound fractions were eluted by gradient with 50 mL of buffer B [50 mM HEPES (pH 8), 500 mM Imidazole, 300 mM NaCl, 5 mM β -Mercaptoethanol, 5% (v/v) Glycerol and EDTA-free protease inhibitor cocktail tablet/50 mL (Roche, UK)]. Peak fractions containing Pol ϵ subunits were pooled and loaded directly onto Strep Trap XT 5 mL (Cytiva) column equilibrated with buffer C [50 mM HEPES (pH 8), 250 mM NaCl, 5 mM β -Mercaptoethanol, 5% (v/v) Glycerol] and EDTA-free protease inhibitor cocktail tablet/50 mL (Roche, UK)]. After loading, the column was washed with 50 mL of buffer C and the bound fractions were incubated for 2-4 hours with 15 mL of 100% buffer D [50 mM HEPES (pH 8), 100 mM NaCl, 50 mM Biotin, 5 mM β -Mercaptoethanol, 5% (v/v) Glycerol] and EDTA-free protease inhibitor cocktail tablet/50 mL (Roche, UK)] and eluted further with 30 mL of 100% buffer D. Fractions that contain all Pol ϵ subunits were pooled and incubated with the TEV protease for overnight and loaded onto an anion exchanger, Mono Q 5/50 GL column (Cytiva) pre-equilibrated with buffer E [50 mM HEPES (pH 8), 100 mM NaCl, 5 mM β -Mercaptoethanol and 5% Glycerol]. After loading, the column was washed with 15 mL of buffer E. The Pol ϵ was eluted with a 10 mL gradient from 100 mM NaCl to 1 M NaCl in 50 mM HEPES (pH 8), 5 mM β -Mercaptoethanol, and 5% Glycerol. The fractions that contained all Pol ϵ subunits were combined and concentrated to 1 mL and loaded onto HiLoad 16/600 Superdex 200 pg (Cytiva) pre-equilibrated with gel filtration buffer [50mM HEPES (8 [DOI](#)), 200 mM NaCl, 1 mM DTT and 5% Glycerol]. Protein fractions were pooled, flash-frozen, and stored at -80°C .

All protein concentrations were calculated by measuring their absorbance at 280 nm using the extinction coefficients calculated from the amino acid composition (CTF18-RFC: $216,360\text{ M}^{-1}\text{ cm}^{-1}$; PCNA: $47,790\text{ M}^{-1}\text{ cm}^{-1}$, Pol ϵ : $351,470\text{ M}^{-1}\text{ cm}^{-1}$).

DNA substrates for Cryo-EM

For the primer/template substrates used in all the structures described here, 80 base pairs template strand 5'-TTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTGGACTGCAC GAATTAAGCAATTCGTAATCATGG TCATAGCT-3' was annealed to a 39 base pairs primer containing a 3' dideoxy cytosine chain terminator 5'-AGCTATG ACCATGATTACGAATTGCTTAATTCGTGCAGT[ddC]-3' to form the P/T substrate. The strands were mixed in an equimolar ratio in the presence of 20 mM Tris pH 7.5 and 25 mM NaCl. The annealing reaction was performed by incubating the P/T at 92°C for 2 minutes followed by slow cooling to room temperature overnight. All oligonucleotides were purchased from IDT Integrated DNA Technologies.

Clamp loading and primer extension substrates

Oligonucleotides for the clamp loading assay comprising the 5'ddPCy3/T P/T DNA substrate were synthesized by Integrated DNA Technologies (Coralville, IA) and purified on denaturing polyacrylamide gels. The concentration of unlabeled template strand DNA was determined from the absorbance at 260 nm using the provided extinction coefficient. Concentrations of the Cy3-labeled primer strand DNA was determined from the extinction coefficient of Cy3 at 550 nm ($\epsilon_{550} = 136,000\text{ M}^{-1}\text{ cm}^{-1}$). For annealing the 5'ddPCy3/T P/T DNA substrate, the Cy3-labeled primer strand DNA and the unlabeled template strand DNA were mixed in equimolar amounts in 1X annealing buffer (10 mM TrisHCl, pH 8.0, 100 mM NaCl, 1 mM EDTA), heated to 95°C for 5 minutes, and allowed to slowly cool to room temperature. For the prime extension assay, Oligo (592:TAACGCCAGGGTTTCCAGTCACG) (Integrated DNA Technologies) was annealed to 50 nM M13mp18 single-stranded DNA (New England Biolabs) in a reaction buffer consisting of [10 mM Tris-HCl (pH 7.6), 100 mM NaCl and 5 mM EDTA]. The mixture was heated to 95°C for 5 minutes, then cooled gradually to room temperature. The unannealed oligonucleotide was removed using a QIAquick PCR purification kit (Qiagen).

Pre-steady state FRET measurements

All experiments were performed at room temperature ($23 \pm 2^\circ\text{C}$) in a 16.100F-Q-10/Z15 sub-micro fluorometer cell (Starna Cells) and monitored in a Horiba Scientific Duetta-Bio fluorescence/absorbance spectrometer as described previously (57). In short, reaction solutions are excited at 514 nm and the fluorescence emission intensities (I) are monitored essentially simultaneously ($\Delta t = 0.118$ ms) at the peak emission wavelengths for Cy3 (563 nm, I_{563}) and Cy5 (665 nm, I_{665}) over time, recording I every 0.17 s. For all experiments, excitation and emission slit widths are 10 nm. All recorded fluorescence emission intensities are corrected by a respective dilution factor and all-time courses are adjusted for the time between the addition of each component and the fluorescence emission intensity recordings (i.e., dead time = 6 s). For any recording of the fluorescence emission intensities (I_{665} and I_{563}), the approximate FRET efficiency is estimated from the equation $E_{\text{FRET}} = \frac{I_{665}}{I_{665} + I_{563}}$. For each experiment below, the final concentrations of all reaction components are indicated.

PCNA loading experiments were carried out in 1X Replication Buffer (25 mM HEPES, pH 7.5, 125 mM KOAc, 10 mM Mg(OAc)₂) supplemented with 1 mM DTT, 1 mM ATP, and the ionic strength adjusted to physiological (200 mM) by addition of KOAc. Experiments were performed via slight modifications to a published protocol (58, 59). In short, 5' ddPCy3/T P/T DNA (200 nM, NeutrAvidin (800 nM homotetramer), and ATP (1 mM) are pre-incubated with RPA (600 nM heterotrimer). Then, Cy5-PCNA (200 nM homotrimer) is added, the resultant solution is transferred to a fluorometer cell, and the cell is placed in the instrument. I_{665} and I_{563} are monitored until both stabilize for at least 1 min. The E_{FRET} values within this stable region represent the signal for Cy5-PCNA remaining completely disengaged from 5' ddPCy3/T•RPA nucleoprotein complex and are averaged to obtain the E_{FRET} value observed prior to addition of a clamp loader complex. Finally, a clamp loader complex (200 nM RFC, CTF18, or CTF18 Δ 165-194-RFC heteropentamer) is added, the resultant solution is mixed, and I_{665} and I_{563} are monitored, beginning 6 s after the addition of the respective clamp loader complex (i.e., $\Delta t = 6$ s).

Primer extension assay

The primer extension reaction was conducted in a reaction buffer containing [40 mM HEPES (pH 7.6), 100 mM K-glutamate, 1 mM DTT, 10 mM Mg-Acetate, 200 $\mu\text{g}/\text{mL}$ BSA, 1 mM ATP, 100 μM each of dATP, dTTP, and dGTP, 20 μM dCTP, and 10 μCi of [³²P]-dCTP (Hartmann)]. Two nanomolar of the primed-M13 template were initially incubated at 37°C for 10 minutes with 100 nM PCNA, 400 nM RPA, and either CTF18-RFCWT or CTF18 Δ 165-194-RFC. The reaction was initiated by the addition of 30 nM Pol ϵ . Reactions were stopped at the indicated time point by adding 50 mM EDTA and electrophoresed on 0.7% alkaline agarose gel at 35 V for 17 hours. The gel was backed with DE81 paper, compressed, and imaged in a Sapphire Biomolecular Imager (Azure Biosystems). All the experiments were performed in triplicate.

Cryo-EM grids preparation and data collection

For all the complexes (WT and mutant), UltraAuFoil® R1.2/1.3 Au 300 grids were glow-discharged at 30 mA for 30 seconds on an EasyGlow glow-discharge unit. For the preparation of the complex without Mg²⁺ (**Dataset 1 and Dataset 3**), P/T DNA, ATP, CTF18-RFC WT/CTF18 Δ 165-194-RFC, and PCNA were mixed in order. The final concentrations were 5.6 μM P/T DNA, 0.2 mM ATP, 1.5 μM CTF18-RFC WT/CTF18 Δ 165-194-RFC, and 7.5 μM PCNA. The buffer this was performed in comprised 50 mM Tris-HCl pH 7.5, 200 mM NaCl, 1 mM DTT, and 0.5 mM ATP. 3 μL of the sample was applied to the grid, blotted for 2 s at blot force -5, and plunged frozen into liquid ethane using a Vitrobot Mark IV (FEI Thermo Fisher), which was set at 4°C and 100% humidity. For the preparation of CTF18-RFC/PCNA/P-T DNA/ATP/Mg²⁺ complex (**Dataset 2**), the components P/T DNA, ATP, Mg²⁺, CTF18-RFC, and PCNA were mixed in order. The final concentrations were 5.6 μM P/T DNA, 0.2 mM ATP, 1.9 mg/mL Mg²⁺, 1.5 μM CTF18-RFC, and 7.5 μM PCNA. The buffer this was performed in

comprised 50 mM Tris-HCl pH 7.5, 200 mM NaCl, 1 mM DTT, 0.5 mM ATP, and 5 mM Mg^{2+} . 3 μ L of the sample was applied to the grid in the same way as above for data collection. Cryo-EM data for all the samples were collected on a Thermo Fisher Scientific Titan Krios G4 transmission electron microscope at KAUST. Electron micrographs for **Dataset 1** (CTF18-RFC-PCNA with ATP in the absence of Mg^{2+}), were collected using a Falcon 4i detector at a dose rate of 7.5 $e^-/\text{pix}/\text{sec}$ for 5 seconds and a calibrated pixel size of 0.93 Å. Focusing was performed over a range between -2.7 μ m and -1.5 μ m, in 0.3 μ m intervals. Electron micrographs for **Dataset 2** (CTF18-RFC-PCNA with ATP and Mg^{2+}) were collected using a Falcon 4i detector at a dose rate of 8.02 $e^-/\text{pix}/\text{sec}$ for 5 seconds and a calibrated pixel size of 0.93 Å. Focusing was performed over a range between -2.7 μ m and -1.5 μ m, in 0.3 μ m intervals. Electron micrographs for **Dataset 3** (CTF18 ^{Δ 165-194}-RFC-PCNA with ATP and in the absence of Mg^{2+}) were collected using a Falcon 4i detector at a dose rate of 7.4 $e^-/\text{pix}/\text{sec}$ for 5 seconds and a calibrated pixel size of 0.93 Å. Focusing was performed over a range between -2.7 μ m and -1.5 μ m, in 0.3 μ m intervals.

Cryo-EM image processing

Preprocessing of all datasets (**Dataset 1**: CTF18-RFC, ATP, primer/template DNA and PCNA; **Dataset 2**: CTF18-RFC, ATP, Mg^{2+} , primer/template DNA and PCNA; **Dataset 3**: CTF18 ^{Δ 165-194}-RFC, ATP, primer/template DNA and PCNA) was performed as follows: the micrographs were corrected for beam-induced motion and then integrated using MotionCor2 ([60](#)). All frames were retained and a patch alignment of 4x4 was used. CTFFIND-4.12 estimated Contrast Transfer Function (CTF) parameters for each micrograph ([61](#)). Integrated movies were inspected with Relion-4.0 (**Dataset 1** and **Dataset 2**) and Relion-5.0 (**Dataset 3**) for further image processing (7906 movies for **Dataset 1**, 6181 movies for **Dataset 2**, and 6499 movies for **Dataset 3**) ([62](#)). Particle picking was performed in automated mode using Topaz ([63](#)) for **Dataset 1**. Particle extraction was carried out from micrographs using a box size of 55 pixels (pixel size: 3.65 Å/pixel). An initial dataset of 2×10^6 particles was cleaned by 2D classification followed by 3D classification with alignment. Three 3D classes were generated with populations of 31, 28, and 40%. 3D refinement, CtfRefine, and polishing were performed on the selected 3D classes corresponding to populations 31 and 40%. A bigger box size, 480 pixels (pixel size: 0.93 Å/pixel) was used to re-extract the particles. 3D refinement, CtfRefine, and polishing yielded reconstruction at 2.9 Å.

Particle picking was performed in reference-based mode for **Dataset 2**. Particle extraction was carried out from micrographs using a box size of 55 pixels (pixel size of 3.65 Å/pixel). An initial dataset of 3×10^6 particles was cleaned by 2D classification followed by 3D classification with alignment. Three 3D classes were generated with 30, 33, and 36% populations. 3D refinement, CtfRefine, and polishing were performed on the selected 3D classes corresponding to populations 33 and 36%. 3D refinement, CtfRefine, and polishing yielded reconstruction at 3.2 Å. Particle picking was performed in automated mode using Topaz ([63](#)) for **Dataset 3**. Particle extraction was done from micrographs using a box size of 220 pixels (pixel size: 0.93 Å/pixel). An initial dataset of 1×10^6 particles was cleaned by 2D classification. For quantification of complex stability, the number of particles contributing to each 2D class was extracted from the classification metadata (**Datasets 1** and **3**). All classes showing isolated PCNA rings were summed and compared to the total number of particles in classes representing intact CTF18-RFC–PCNA complexes. This analysis was performed for both wild-type and β -hairpin deletion mutant datasets. Notably, no 2D classes corresponding to free PCNA were observed in the wild-type dataset, whereas in the mutant dataset, a substantial fraction of particles corresponded to isolated PCNA, suggesting reduced stability of the mutant complex.

Cryo-EM 3D Variability Analysis

3DVA ([43](#), [44](#)) of the CTF18-RFC–PCNA consensus maps, with and without Mg^{2+} , was performed in CryoSPARC v 4.6 using their respective refinement masks, employing three modes and a resolution filter of 5 Å, based on global FSC resolutions of 2.9 Å and 3.2 Å, respectively. For

each mode, 3 frames were generated and morphed to visualize the conformational flexibility across the components.

Molecular Modelling

The structure of the Ctf18 large subunit of human CTF18-RFC (Q8WVB6, residues 277-805) was generated with Phyre2 ([64](#)), using as a template the sequence of human RFC1 (P35251, residues 581-1034). The Ctf18 subunit model was then aligned to the RFC1 subunit in the structure of the human RFC clamp loader bound to PCNA in an autoinhibited conformation (PDB 6VVO, ([10](#))) using Chimera, the RFC1 subunit was deleted, and the resulting model including Ctf18, subunits RFC2-5 and PCNA was rigid-body fitted into the cryo-EM map. The model was edited and real-space refined with Coot ([65](#)) following flexible fitting performed with ISOLDE ([66](#)). A final real-space refinement was performed in Phenix ([67](#)), applying secondary structure constraints. The models of both CTF18-RFC-PCNA complexes with and without the Mg^{2+} ion were generated similarly.

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Additional information

Data Availability

The maps of CTF18-RFC-ATP-PCNA and CTF18-RFC-ATP- Mg^{2+} -PCNA complexes have been deposited in the EMBD with accession codes EMD-60534, and EMD-60598 and the atomic models in the Protein Data Bank under accession codes PDB 8ZWO and PDB 9IIN.

Author Contributions

S.M.H. and A.D.B. conceived and supervised the study. M.T. and A.A. prepared proteins, carried out primer extension assays and analyzed the data. G.Y. and M.H. prepared proteins, carried out clamp loading assays and analyzed the data. G.R.B. and C.G.S. performed cryo-EM imaging and processing. A.U.D. performed the cryo-EM 3DVA analysis. G.R.B., P.Q-N., and A.D.B. performed model building and validation. A.D.B. wrote the manuscript with contributions from the other authors.

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Additional files

Supplementary Information File [↗](#)

Supplementary Video 1 [↗](#)

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Reviewer #1 (Public review):

Summary:

The authors report the structure of the human CTF18-RFC complex bound to PCNA. Similar structures (and more) have been reported by the O'Donnell and Li labs. This study should add to our understanding of CTF18-RFC in DNA replication and clamp loaders in general. However, there are numerous major issues that I recommend the authors fix.

Strengths:

The structures reported are strong and useful for comparison with other clamp loader structures that have been reported lately.

Comments on revisions:

The revised manuscript is greatly improved. The comparison with hRFC and the addition of direct PCNA loading data from the Hedglin group are particular highlights. I think this is a strong addition to the literature.

I only have minor comments on the revised manuscript.

(1) The clamp loading kinetic data in Figure 6 would be more easily interpreted if the three graphs all had the same x axes, and if addition of RFC was $t=0$ rather than $t=60$ sec.

(2) The author's statement that "CTF18-RFC displayed a slightly faster rate than RFC" seems to me a bit misleading, even though this is technically correct. The two loaders have indistinguishable rate constants for the fast phase, and RFC is a bit slower than CTF18-RFC in the slow phase. However, the data also show that RFC is overall more efficient than CTF18-RFC at loading PCNA because much more flux through the fast phase (rel amplitudes 0.73 vs 0.36). Because the slow phase represents such a reduced fraction of loading events, the slight reduction in rate constant for the slow phase doesn't impact RFC's overall loading. And because the majority of loading events are in the fast phase, RFC has a faster halftime than CTF18-RFC. (Is it known what the different phases correspond to? If it is known, it might be interesting to discuss.)

(3) AAA+ is an acronym for "ATPases Associated with diverse cellular Activities" rather than "Adenosine Triphosphatase Associated".

<https://doi.org/10.7554/eLife.103493.2.sa3>

Reviewer #2 (Public review):

Summary

Briola and co-authors have performed a structural analysis of the human CTF18 clamp loader bound to PCNA. The authors purified the complexes and formed a complex in solution. They used cryo-EM to determine the structure to high resolution. The complex assumed an auto-inhibited conformation, where DNA binding is blocked, which is of regulatory importance and suggests that additional factors could be required to support PCNA loading on DNA. The authors carefully analysed the structure and compared it to RFC and related structures.

Strength & Weakness

Their overall analysis is of high quality, and they identified, among other things, a human-specific beta-hairpin in Ctf18 that flexible tethers Ctf18 to Rfc2-5. Indeed, deletion of the beta-hairpin resulted in reduced complex stability and a reduction in a primer extension assay

with Pol ϵ . Moreover, the authors identify that the Ctf18 ATP-binding domain assumes a more flexible organisation.

The data are discussed accurately and relevantly, which provides an important framework for rationalising the results.

All in all, this is a high-quality manuscript that identifies a key intermediate in CTF18-dependent clamp loading.

Comments on revisions:

The authors have done a nice job with the revision.

<https://doi.org/10.7554/eLife.103493.2.sa2>

Reviewer #3 (Public review):

Summary:

CTF18-RFC is an alternative eukaryotic PCNA sliding clamp loader which is thought to specialize in loading PCNA on the leading strand. Eukaryotic clamp loaders (RFC complexes) have an interchangeable large subunit which is responsible for their specialized functions. The authors show that the CTF18 large subunit has several features responsible for its weaker PCNA loading activity, and that the resulting weakened stability of the complex is compensated by a novel beta hairpin backside hook. The authors show this hook is required for the optimal stability and activity of the complex.

Relevance:

The structural findings are important for understanding RFC enzymology and novel ways that the widespread class of AAA ATPases can be adapted to specialized functions. A better understanding of CTF18-RFC function will also provide clarity into aspects of DNA replication, cohesion establishment and the DNA damage response.

Strengths:

The cryo-EM structures are of high quality enabling accurate modelling of the complex and providing a strong basis for analyzing differences and similarities with other RFC complexes.

Weaknesses:

The manuscript would have benefited from a more detailed biochemical analysis using mutagenesis and assays to tease apart the differences with the canonical RFC complex. Analysis of the FRET assay could be improved.

Overall appraisal:

Overall, the work presented here is solid and important. The data is mostly sufficient to support the stated conclusions.

Comments on revisions:

While the authors addressed my previous specific concerns, they have now added a new experiment which raises new concerns.

The FRET clamp loading experiments (Fig. 6) appear to be overfitted so that the fitted values are unlikely to be robust and it is difficult to know what they mean, and this is not explained in this manuscript. Specifically, the contribution of two exponentials is floated in each

experiment. By eye, CTF18-RFC looks much slower than RFC1-RFC (as also shown previously in the literature) but the kinetic constants and text suggest it is faster. This is because the contribution of the fast exponential is substantially decreased, and the rate constants then compensate for this. There is a similar change in contribution of the slow and fast rates between WT CTF18 and the variant (where the data curves look the same) and this has been balanced out by a change in the rate constants, which is then interpreted as a defect. I doubt the data are strong enough to confidently fit all these co-dependent parameters, especially for CTF18, where a fast initial phase is not visible. I would recommend either removing this figure or doing a more careful and thorough analysis.

<https://doi.org/10.7554/eLife.103493.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

Summary:

The authors report the structure of the human CTF18-RFC complex bound to PCNA. Similar structures (and more) have been reported by the O'Donnell and Li labs. This study should add to our understanding of CTF18-RFC in DNA replication and clamp loaders in general. However, there are numerous major issues that I recommend the authors fix.

Strengths:

The structures reported are strong and useful for comparison with other clamp loader structures that have been reported lately.

Weaknesses:

The structures don't show how CTF18-RFC opens or loads PCNA. There are recent structures from other groups that do examine these steps in more detail, although this does not really dampen this reviewer's enthusiasm. It does mean that the authors should spend their time investigating aspects of CTF18-RFC function that were overlooked or not explored in detail in the competing papers. The paper poorly describes the interactions of CTF18-RFC with PCNA and the ATPase active sites, which are the main interest points. The nomenclature choices made by the authors make the manuscript very difficult to read.

Reviewer #2 (Public review):

Summary

Briola and co-authors have performed a structural analysis of the human CTF18 clamp loader bound to PCNA. The authors purified the complexes and formed a complex in solution. They used cryo-EM to determine the structure to high resolution. The complex assumed an auto-inhibited conformation, where DNA binding is blocked, which is of regulatory importance and suggests that additional factors could be required to support PCNA loading on DNA. The authors carefully analysed the structure and compared it to RFC and related structures.

Strength & Weakness

Their overall analysis is of high quality, and they identified, among other things, a human-specific beta-hairpin in Ctf18 that flexibly tethers Ctf18 to Rfc2-5. Indeed, deletion of the beta-hairpin resulted in reduced complex stability and a reduction in a primer

extension assay with Pol ϵ . This is potentially very interesting, although some more work is needed on the quantification. Moreover, the authors argue that the Ctf18 ATP-binding domain assumes a more flexible organisation, but their visual representation could be improved.

The data are discussed accurately and relevantly, which provides an important framework for rationalising the results.

All in all, this is a high-quality manuscript that identifies a key intermediate in CTF18-dependent clamp loading.

Reviewer #3 (Public review):

Summary:

CTF18-RFC is an alternative eukaryotic PCNA sliding clamp loader that is thought to specialize in loading PCNA on the leading strand. Eukaryotic clamp loaders (RFC complexes) have an interchangeable large subunit that is responsible for their specialized functions. The authors show that the CTF18 large subunit has several features responsible for its weaker PCNA loading activity and that the resulting weakened stability of the complex is compensated by a novel beta hairpin backside hook. The authors show this hook is required for the optimal stability and activity of the complex.

Relevance:

The structural findings are important for understanding RFC enzymology and novel ways that the widespread class of AAA ATPases can be adapted to specialized functions. A better understanding of CTF18-RFC function will also provide clarity into aspects of DNA replication, cohesion establishment, and the DNA damage response.

Strengths:

The cryo-EM structures are of high quality enabling accurate modelling of the complex and providing a strong basis for analyzing differences and similarities with other RFC complexes.

Weaknesses:

The manuscript would have benefitted from more detailed biochemical analysis to tease apart the differences with the canonical RFC complex.

I'm not aware of using Mg depletion to trap active states of AAA ATPases. Perhaps the authors could provide a reference to successful examples of this and explain why they chose not to use the more standard practice in the field of using ATP analogues to increase the lifespan of reaction intermediates.

Overall appraisal:

Overall the work presented here is solid and important. The data is sufficient to support the stated conclusions and so I do not suggest any additional experiments.

Reviewer #1 (Recommendations for the authors):

We thank the reviewer for their positive comments and for their thorough review. All raised points have been addressed below.

Major points

(1) The nomenclature used in the paper is very confusing and sometimes incorrect. The authors refer to CTF18 protein as "Ctf18", and the entire CTF18-RFC complex as "CTF18". This results in massive confusion because it is hard to ascertain whether the authors are discussing the individual subunits or the entire complex. Because these are human proteins, each protein name should be fully capitalized (i.e. CTF18, RFC4 etc). The full complex should be referred to more clearly with the designation CTF18-RFC or CTF18-RLC (RFC-like complex). Also, because the yeast and human clamp loader complexes use the same nomenclature for different subunits, it would be best for the authors to use the "A, B, C, D, E subunit" nomenclature that has been standard in the field for the past 20 years. Finally, the authors try to distinguish PCNA subunits by labeling them "PCNA2" or "PCNA1" (see Page 8 lines 180,181 for an example). This is confusing because the names of the RFC subunits have similar formats (RFC2, RFC3, RFC4, etc). In the case of RFC this denotes unique genes, whereas PCNA is a homotrimer. Could the authors think of another way to denote the different subunits, such as super/subscript? PCNA-I, PCNA-II, PCNA-III?

We thank the reviewer for pointing out the confusing nomenclature. Following the referee suggestion, we now refer to the CTF18 full complex as "CTF18-RFC". We prefer keeping the nomenclature used for CTF18 subunits as RFC2, RFC3 etc., as recently used in Yuan et al, Science, 2024. However, we followed the referee's suggestion for PCNA subunits, now referred to as PCNA-I, PCNA-II and PCNA-III.

(2) I believe that the authors are over-interpreting their data in Figure 1. The claim that "less sharp definition" of the map corresponding to the AAA+ domain of Ctf18 supports a relatively high mobility of this subunit is largely unsubstantiated. There are several reasons why one could get varying resolution in a cryo-EM reconstruction, such as compositional heterogeneity, preferred orientation artifacts, or how the complex interacts with the air-water interface. If other data were presented that showed this subunit is flexible, this evidence would support that data but cannot alone as justification for subunit mobility. Along these lines, how was the buried surface area (2300 vs 1400 Å²) calculated? Is this the total surface area or only the buried surface area involving the AAA+ domains? It is surprising that these numbers are so different considering that the subunits and complexes look so similar (Figures 1c and 2b).

We respectfully disagree with the suggestion that our interpretation of local flexibility in the AAA+ domain of Ctf18 is overreaching. Several lines of evidence support this interpretation. First, compositional heterogeneity is unlikely, as the A' domain of Ctf18 is well-resolved and forms stable interactions with RFC3, indicating that Ctf18 is consistently incorporated into the complex. Second, preferred orientation artifacts are excluded, as the particle distribution shows excellent angular coverage (Fig. S9a). Third, we now include a 3D variability analysis (3DVA; Supplementary Video 1), which reveals local conformational heterogeneity centered around the AAA+ domain of Ctf18, consistent with intrinsic flexibility.

Regarding the buried surface area values, the reported numbers refer specifically to the interfaces between the AAA+ domain of Ctf18 and RFC2, and are derived from buried surface area calculations performed with PISA. The smaller interface (~1400 Å²) compared to RFC1–RFC2 (~2300 Å²) reflects low sequence identity (~26%) and divergent structural features, including the absence of conserved elements such as the canonical PIP-box in Ctf18. We have clarified and expanded this explanation in the revised manuscript (Page 7).

(3) The authors very briefly discuss interactions with PCNA and how the CTF18-RFC complex differs from the RFC complex. This is amongst the most interesting results from their work, but also not well-developed. Moreover, Figure 3D describing these

interactions is extremely unclear. I feel like this observation had potential to be interesting, but is largely ignored by the authors.

We thank the referee for pointing this out. We have expanded the section describing the interactions of CTF18-RFC and PCNA (Page 9 in the new manuscript), and made a new panel figure with further details (Fig. 3D).

(4) The authors make the observation that key ATP-binding residues in RFC4 are displaced and incompatible with nucleotide binding in their CTF18-RFC structure compared to the hRFC structure. This should be a main-text figure showing these displacements and how it is incompatible with ATP binding. Again, this is likely an interesting finding that is largely glossed over by the authors.

We now discuss this feature in detail (Pag 11 in the new manuscript), and added two figure insets (Fig. 4c) describing the incompatibility of RFC4 with nucleotide binding.

(5) The authors claim that the work of another group (citation 50) "validate(s) our predictions regarding the significant similarities between CTF18-RFC and canonical RFC in loading PCNA onto a ss/dsDNA junction." However, as far as this reviewer can tell the work in citation 50 was posted online before the first draft of this manuscript appeared on biorxiv, so it is dubious to claim that these were "predictions."

We agree with the referee about this claim. We have now revised the text as follows:

“While our work was being finalized, several cryo-EM structures of human CTF18-RFC bound to PCNA and primer/template DNA were reported by another group (He et al, PNAS, 2024). These findings are consistent with the distinct features of CTF18-RFC observed in our structures and independently support the notion of significant mechanistic similarity between CTF18-RFC and canonical RFC in loading PCNA onto a ss/dsDNA junction”.

(6) The authors use a primer extension assay to test the effects of truncating the Nterminal beta hairpin of CTF18. However, this assay is only a proxy for loading efficiency and the observed effects of the mutation are rather subtle. The authors could test their hypothesis more clearly if they performed an ATPase assay or even better a clamp loading assay.

We thank the referee for this valuable suggestion. In response, we have performed clamp loading assays comparing the activities of human RFC, wild-type CTF18-RFC, and the β -hairpin-truncated CTF18-RFC mutant. The results, now presented in Fig. 6 and Table 1 of the revised manuscript, clearly show that truncation of the N-terminal β hairpin results in a slower rate of PCNA loading. We propose that this reduced loading rate likely contributes to the diminished Pol ϵ -mediated DNA synthesis observed in the primer extension assays.

Minor points

(1) Page 3 line 53 the introduction suggests that ATP hydrolysis prompts clamp closure. While this may be the case, to my knowledge all recent structural work shows that closure can occur without ATP hydrolysis. It may be better to rephrase it to highlight that under normal loading conditions, ATP hydrolysis occurs before clamp closure.

The text now reads (Page 3):

“DNA binding prompts the closure of the clamp and hydrolysis of ATP induces the concurrent disassembly of the closed clamp loader from the sliding clamp-DNA complex, completing the cycle necessary for the engagement of the replicative polymerases to start DNA synthesis.”

(2) Page 3 line 60, I do not see how the employment of alternative loaders highlights the specificity of the loading mechanism - would it not be possible for multiple loaders to have promiscuous clamp loading?

We thank the referee for this comment. The text now reads (Page 3):

“However, eukaryotes also employ alternative loaders (20), including CTF18-RFC (6, 21-24), which likely use a conserved loading mechanism but are functionally specialized through specific protein interactions and context-dependent roles in DNA replication.”

(3) Page 4 line 75 could you please cite a study that shows Ctf8 and Dcc1 bind to the Ctf18 C-terminus and that a long linker is predicted to be flexible?

Two references have been added (Stokes et al, NAR, 2020 and Grabarczyk et al, Structure, 2018)

(4) Figure 2A has the N-terminal region of Ctf18 as bound to RFC3 but should likely be labeled as bound to RFC5. This caused significant confusion while trying to parse this figure. Further, the inclusion of "X" as a sequence - does this refer to a sequence that was not buildable in the cryo-EM map? I would be surprised that density immediately after the conserved DEXX box motif is unbuildable. If this is the case, it should be clearly stated in the figure legend that "X" denotes an unbuildable sequence. For the conserved beta-hairpin in the sequence, could the authors superimpose the AlphaFold prediction onto their structure? It would be more informative than just looking at the sequence.

We apologize for this confusion. The error in Figure 2A has been corrected. The figure caption now explicitly says that “X” refers to amino acid residues in the sequence which were not modelled. A superposition of the cryo-EM model of the N-terminal Beta hairpin in human Ctf18 and AlphaFold predictions for this feature in drosophila and yeast Ctf18 is now presented in Figure 2A.

(5) Page 8 line 168, the use of the term "RFC5" here feels improper, since the "C" subunit is not RFC5 in all lower eukaryotes (see comment above about nomenclature). For instance, in S cerevisiae, the C subunit is RFC3. I would expect this interaction to be maintained in all C subunits, not all RFC5 subunits.

The text now reads (Page 8):

“Therefore, lower eukaryotes may use a similar b-hairpin motif to bind the corresponding subunit of the RFC-module complex (RFC5 in human, Rfc3 in S. cerevisiae), emphasizing its importance.”

(6) Page 10 line 228, the authors claim that hydrolysis is dispensable at the Ctf18/RFC2 interface based on evidence from RFC1/RFC2 interface, by analogy that this is the "A/B" interface in both loaders. However, the wording makes it sound as if the cited data were collected while studying Ctf18 loaders. The authors should clarify this point.

The text has been modified as follows (Pag 11):

“Prior research has indicated that hydrolysis at the large subunit/RFC2 interface is not essential for clamp loading by various loaders (48-51), while the others are critical for the clamp-loading activity of eukaryotic RFCs. “

(7) Page 11 line 243/244 the authors introduce the separation pin. Could they clarify whether Ctf18 contains any aromatic residues in this structural motif that would suggest it serves the same functional purpose? Also, the authors highlight this is similar to yeast RFC, which makes it sound like this is not conserved in human RFC, but the structural motif is also conserved in human RFC.

We thank the reviewer for this helpful comment. We have clarified in the revised text (Page 12) that the separation pin is conserved not only in yeast RFC but also in human RFC, and now note that human Ctf18 also harbors aromatic residues at the corresponding positions. This observation is supported by the new panel in Figure 4e.

Minutia

(1) Page 2 line 37 please remove the word "and" before PCNA.

This has been corrected.

(2) Please define AAA+ and update the language to clarify that not all pentameric AAA+ ATPases are clamp loaders.

AAA+ has been now defined (Page 3).

(3) Page 4 line 86 Given the relatively weak interaction of Pol ϵ .

This has been corrected.

(4) Page 8 line 204 the authors likely mean "leucine" and not "lysine".

We thank the reviewer for catching this. The error has been corrected.

(5) Page 14 line 300, the authors claim that CTF18 utilizes three subunits but then list four.

We have corrected this.

Reviewer #2 (Recommendations for the authors):

We thank the reviewer for their positive comments and valuable suggestions. The points raised by the referee have been addressed below.

Major point:

(1) Please quantify Figure 6 and S9 from 3 independent repeats and determine the standard deviation to show the variability of the Ctf18 beta hairpin deletion. The authors suggest that a suboptimal Ctf18 complex interaction with PCNA impacts the stability of the complex, but do not test this hypothesis. Could the suboptimal PIP motif in Ctf18 be changed to an improved motif and the impact tested in the primer extension assay? Although not essential, it would be a nice way to explore the mechanism.

We thank the reviewer for the suggestion. However, we note that Figure 6b (now 7b) already presents the quantification of the primer extension assay from three independent replicates, with error bars showing standard deviations, and includes the calculated rate of product accumulation. These data clearly indicate a 42% reduction in primer synthesis rate upon deletion of the Ctf18 β -hairpin.

We agree that we do not provide direct evidence of impaired complex stability upon deletion of the Ctf18 β -hairpin. However, the 2D classification of the cryo-EM dataset (Figure S9) shows a marked reduction in the number of particles corresponding to intact CTF18-RFC-PCNA complexes in the β -hairpin deletion sample, with the majority of particles corresponding to free PCNA. This contrasts with the wild-type dataset, where complex particles are predominant. These findings indirectly suggest that deletion of the β -hairpin compromises the stability or assembly of the clamp-loader-clamp complex.

We thank the reviewer for the valuable suggestion to mutate the weak PIP-box of Ctf18. While an interesting direction, we instead sought to directly test the mechanism by performing quantitative clamp loading assays. These assays revealed a significant reduction in the rate of PCNA loading by the CTF18 Δ 165–194-RFCmutant (Figure 6), supporting the conclusion that the β -hairpin contributes to productive PCNA loading. This loading delay likely underlies the reduced rate of primer extension observed in the Pol ϵ assay (Figure 7), consistent with impaired formation of processive polymerase– clamp complexes.

(2) I did not see the method describing how the 2D classes were quantified to evaluate the impact of the Ctf18 beta hairpin deletion on complex formation. Please add the relevant information.

The relevant information has been added to the Method section:

“For quantification of complex stability, the number of particles contributing to each 2D class was extracted from the classification metadata (Datasets 1 and 3). All classes showing isolated PCNA rings were summed and compared to the total number of particles in classes representing intact CTF18-RFC-PCNA complexes. This analysis was performed for both wild-type and β -hairpin deletion mutant datasets. Notably, no 2D classes corresponding to free PCNA were observed in the wild-type dataset, whereas in the mutant dataset, a substantial fraction of particles corresponded to isolated PCNA, suggesting reduced stability of the mutant complex.”

Minor point:

(1) Page 2, line 25. Detail what type of mobility is referred to. Do you mean flexibility in the EM-map?

We have clarified this. The text now reads:

“The unique RFC1 (Ctf18) large subunit of CTF18-RFC, which based on the cryo-EM map shows high relative flexibility, is anchored to PCNA through an atypical low-affinity PIP box”

(2) Page 4, line 82. Please introduce CMGE, or at least state what the abbreviation stands for.

This has been addressed.

(3) Page 4, line 89. Specify that the architecture of the HUMAN CTF18-RFC module is not known, as the yeast one has been published.

At the time our study was initiated, the architecture of the human CTF18-RFC module was unknown. A structure of the human complex was published by another group during the final stages of our work and is now properly acknowledged in the Discussion.

(4) Page 6. Is it possible to illustrate why the autoinhibited state cannot bind to DNA? A visual representation would be nice.

We thank the reviewer for this suggestion. Figure 4b in the original manuscript already illustrates why the autoinhibited, overtwisted conformation of the CTF18-RFC pentamer cannot accommodate DNA. In this state, the inner chamber of the loader is sterically occluded, precluding the binding of duplex DNA.

Reviewer #3 (Recommendations for the authors):

We thank Reviewer #3 for their constructive feedback and positive overall assessment of our work.

We also thank the reviewer for their remarks on the use of Mg depletion to halt hydrolysis. Magnesium is an essential cofactor for ATP hydrolysis, and its depletion is expected to effectively prevent catalysis by destabilizing the transition state, possibly more completely than the use of slowly hydrolysable analogues such as ATP γ S. We have recently employed Mg²⁺ depletion to successfully trap a pre-hydrolytic intermediate in a replicative AAA+ helicase engaged in DNA unwinding (Shahid et al., Nature, 2025). This precedent supports the rationale for our choice, and the reference has now been included in the revised manuscript.

I think the authors deposited the FSC curve for the +Mg structure in the -Mg structure PDB/EMDB entry according to the validation report.

We thank the reviewer for their careful inspection of the deposition materials. The discrepancy in the deposited FSC curve has now been corrected, and the appropriate FSC curves have been assigned to the correct PDB/EMDB entries.

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